# 1 Functions and Models

# 1.1 Four Ways to Represent a Function

# Representations of Functions

We consider four different ways to represent a function:

- verbally (by a description in words)
- numerically (by a table of values)
- visually (by a graph)
- algebraically (by an explicit formula)

# Four Ways to Represent a Function

Functions arise whenever one quantity depends on another. Consider the following four situations.

**A.** The area 面积 $A$ of a circle depends on the radius 半径 $r$ of the circle. The rule that connects $r$ and $A$ is given by the equation 方程 $A = \pi r^2$. With each positive 正数 number $r$ there is associated one value of $A$, and we say that $A$ is a *function* of $r$.

# Four Ways to Represent a Function

**B.** The human population of the world *P* depends on the time *t*. The 表格 table gives 估记 estimates of the world population $P(t)$ at time *t*, for certain years. For instance,

$$P(1950) \approx 2{,}560{,}000{,}000$$

But for each 值 value of the time *t* there is a 相应的 corresponding valu of *P*, and we say that *P* is a function of *t*.

| Year | Population (millions) |
|---|---|
| 1900 | 1650 |
| 1910 | 1750 |
| 1920 | 1860 |
| 1930 | 2070 |
| 1940 | 2300 |
| 1950 | 2560 |
| 1960 | 3040 |
| 1970 | 3710 |
| 1980 | 4450 |
| 1990 | 5280 |
| 2000 | 6080 |
| 2010 | 6870 |

# Four Ways to Represent a Function

**C.** The cost *C* of shipping a package depends on its weight *w*. Although there is no simple formula that connects *w* and *C*, the post office has a rule for determining *C* when *w* is known.

# Four Ways to Represent a Function

A **function** $f$ is a rule that assigns to each element $x$ in a set $D$ exactly one element, called $f(x)$, in a set $E$.

We usually consider functions for which the sets *D* and *E* are sets of real numbers. The set *D* is called the **domain** of the function. domain 定义域

The number $f(x)$ is the **value of $f$ at $x$** and is read "$f$ of $x$." The **range** of $f$ is the set of all possible values of $f(x)$ as $x$ varies throughout the domain. range 值域

A symbol that represents an arbitrary number in the *domain* of a function $f$ is called an **independent variable**.

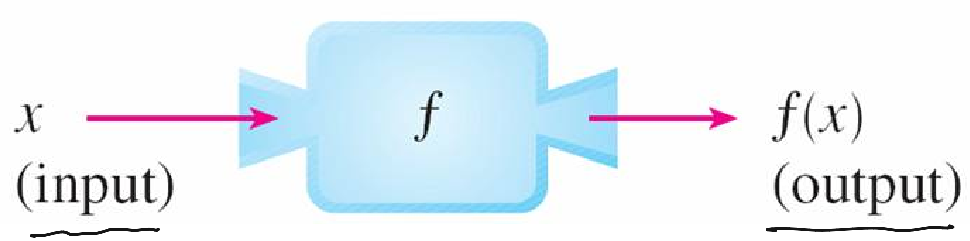
independent variable 自变量

# Four Ways to Represent a Function

dependent variable 因变量

A symbol that represents a number in the *range* of $f$ is called a **dependent variable**. In Example A, for instance, $r$ is the independent variable and $A$ is the dependent variable.

It's helpful to think of a function as a **machine** (see Figure 2).




Machine diagram for a function $f$

**Figure 2**

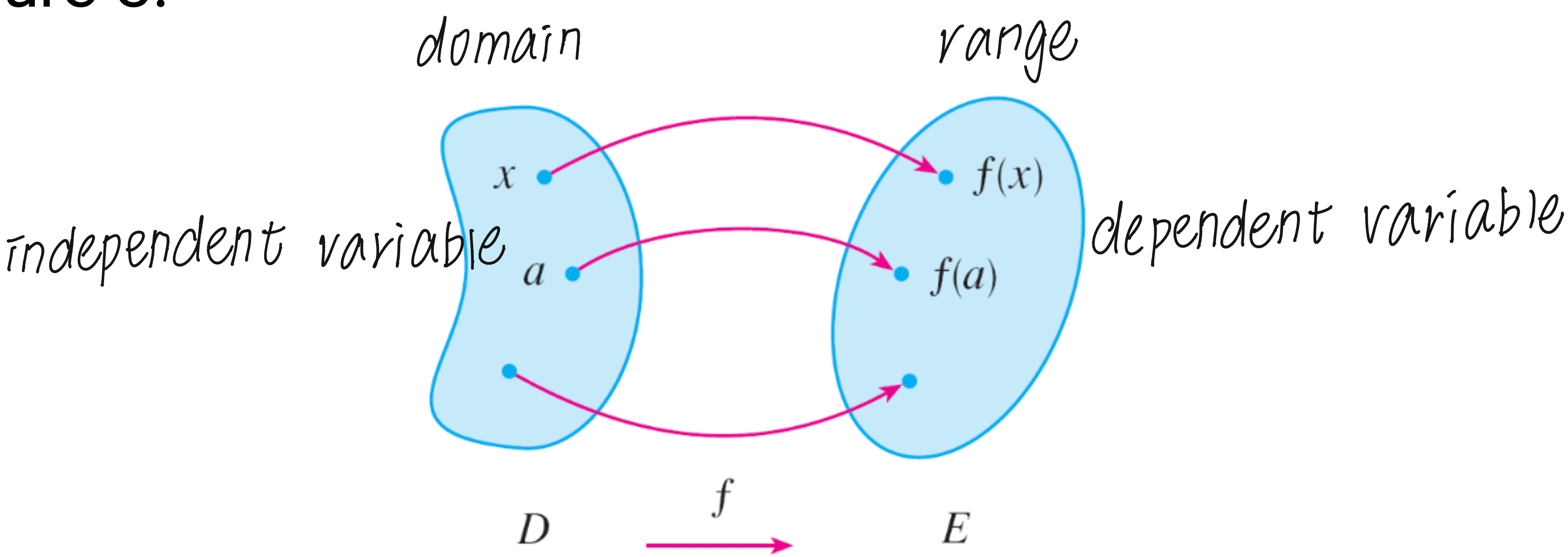
# Four Ways to Represent a Function

If $x$ is in the domain of the function $f$, then when $x$ enters the machine, it's accepted as an input and the machine produces an output $f(x)$ according to the rule of the function.

Thus we can think of the domain as the set of all possible inputs and the range as the set of all possible outputs.

# Four Ways to Represent a Function

Another way to picture a function is by an **arrow diagram** as in Figure 3.




Arrow diagram for *f*

**Figure 3**

Each arrow connects an element of *D* to an element of *E*. The arrow indicates that $f(x)$ is associated with $x$, $f(a)$ is associated with $a$, and so on.

# Four Ways to Represent a Function

The most common method for visualizing a function is its graph. If $f$ is a function with domain $D$, then its **graph** is the set of ordered pairs

$$\{(x, f(x)) \mid x \in D\}$$

In other words, the graph of $f$ consists of all points $(x, y)$ in the coordinate plane such that $y = f(x)$ and $x$ is in the domain of $f$.

The graph of a function $f$ gives us a useful picture of the behavior or "life history" of a function.

# Four Ways to Represent a Function

Since the $y$-coordinate of any point $(x, y)$ on the graph is $y = f(x)$, we can read the value of $f(x)$ from the graph as being the height of the graph above the point $x$ (see Figure 4).

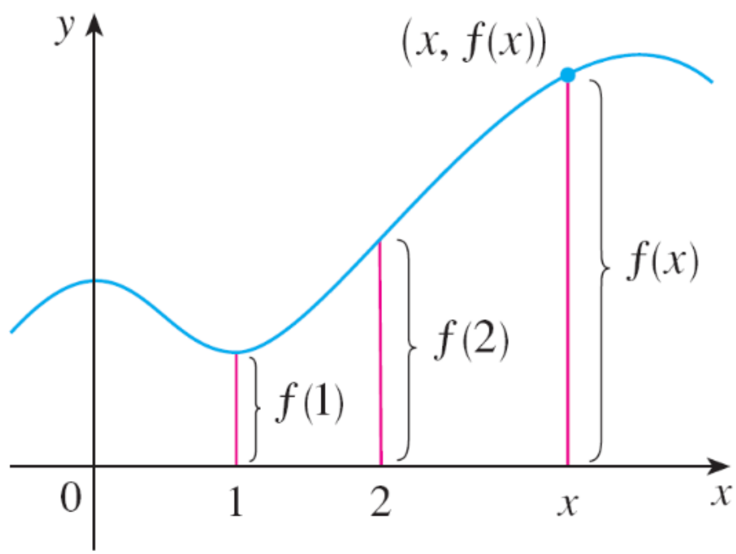



**Figure 4**

# Four Ways to Represent a Function

The graph of $f$ also allows us to picture the domain of $f$ on the $x$-axis and its range on the $y$-axis as in Figure 5.

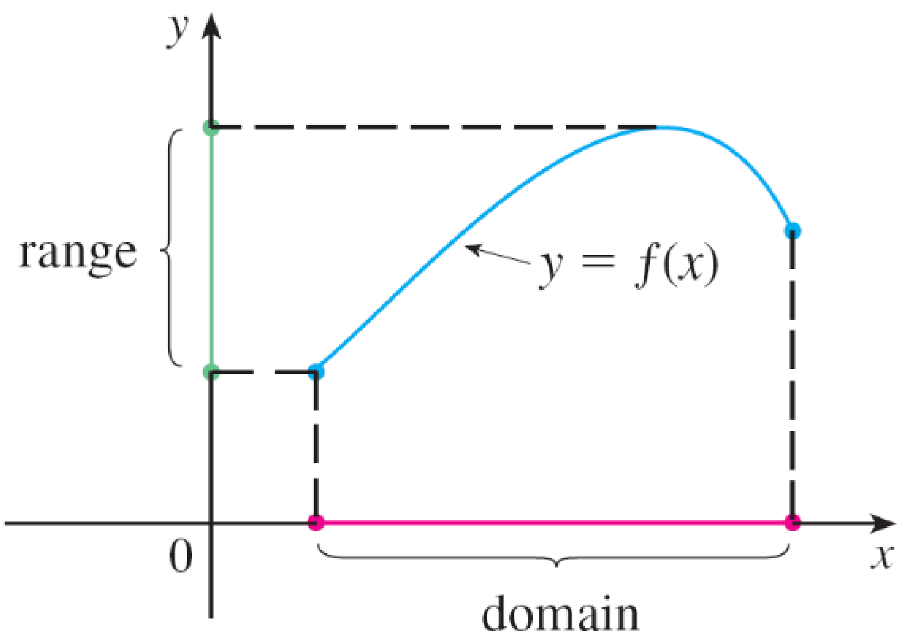



**Figure 5**

# Example 1

The graph of a function $f$ is shown in Figure 6.

**(a)** Find the values of $f(1)$ and $f(5)$.

**(b)** What are the domain and range of $f$?

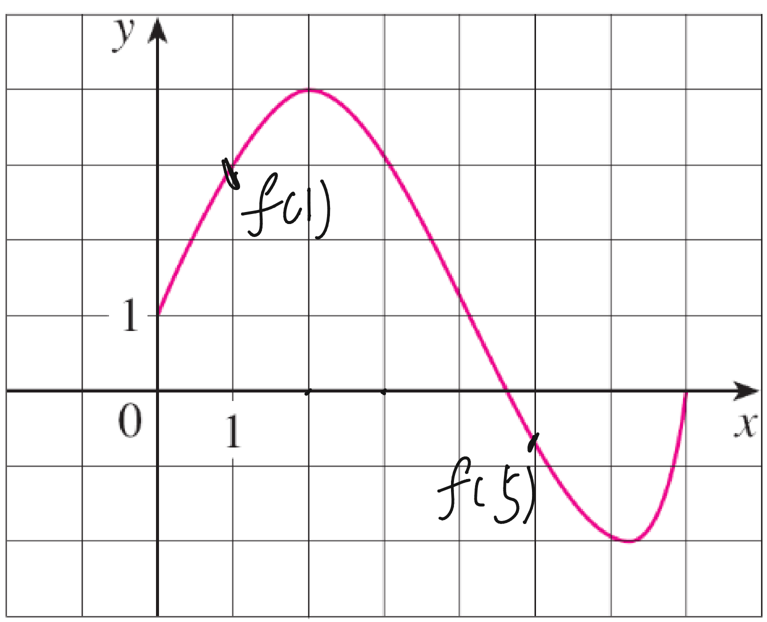



Figure 6

# Example 1 – *Solution*

**(a)** We see from Figure 6 that the point (1, 3) lies on the graph of $f$, so the value of $f$ at 1 is $f(1) = 3$. (In other words, the point on the graph that lies above $x = 1$ is 3 units above the $x$-axis.)

When $x = 5$, the graph lies about 0.7 unit below the $x$-axis, so we estimate that $f(5) \approx -0.7$.

**(b)** We see that $f(x)$ is defined when $0 \leq x \leq 7$, so the domain of $f$ is the closed interval [0, 7]. Notice that $f$ takes on all values from –2 to 4, so the range of $f$ is

$$\{y \,/\, -2 \leq y \leq 4\} = [-2, 4]$$

# Representations of Functions

The graph of a function is a curve in the *xy*-plane.

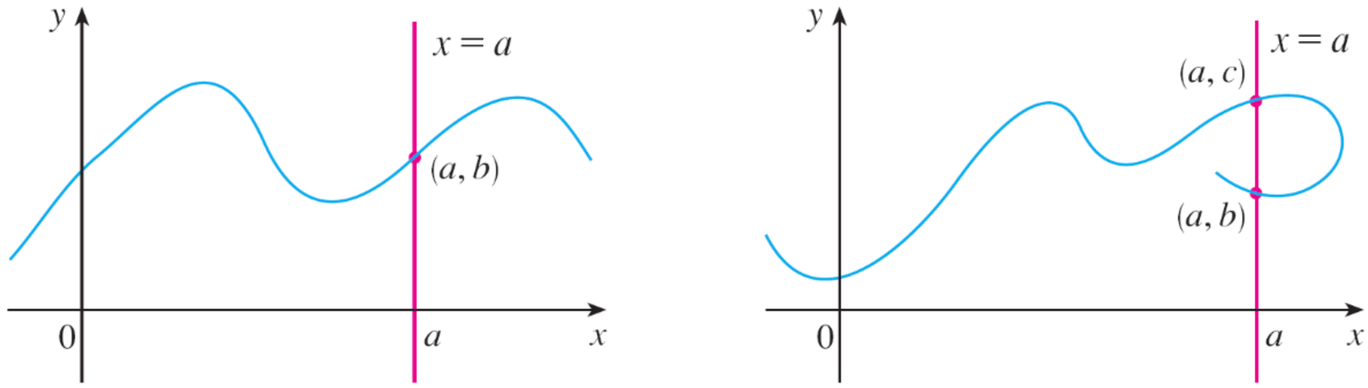
But the question arises: Which curves in the *xy*-plane are graphs of functions?

This is answered by the following test.

**The Vertical Line Test** A curve in the *xy*-plane is the graph of a function of $x$ if and only if no vertical line intersects the curve more than once.

# Representations of Functions

The reason for the truth of the Vertical Line Test can be seen in Figure 13.




(a) This curve represents a function.

(b) This curve doesn't represent a function.

**Figure 13**

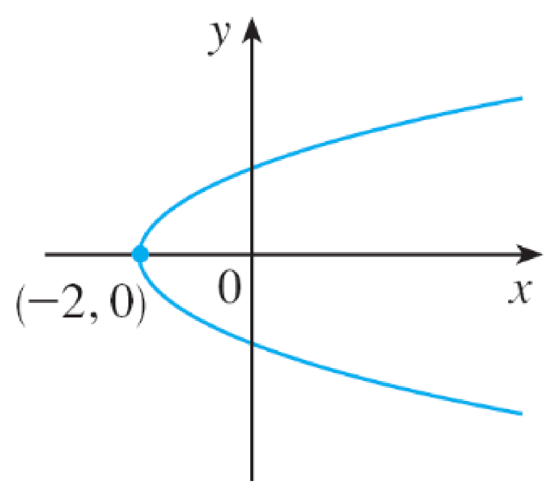
# Representations of Functions

If each vertical line $x = a$ intersects a curve only once, at $(a, b)$, then exactly one functional value is defined by $f(a) = b$.

But if a line $x = a$ intersects the curve twice, at $(a, b)$ and $(a, c)$, then the curve can't represent a function because a function can't assign two different values to $a$.

# Representations of Functions

For example, the parabola $x = y^2 - 2$ shown in Figure 14(a) is not the graph of a function of $x$ because, as you can see, there are vertical lines that intersect the parabola twice. The parabola, however, does contain the graphs of *two* functions of $x$.




$x = y^2 - 2$

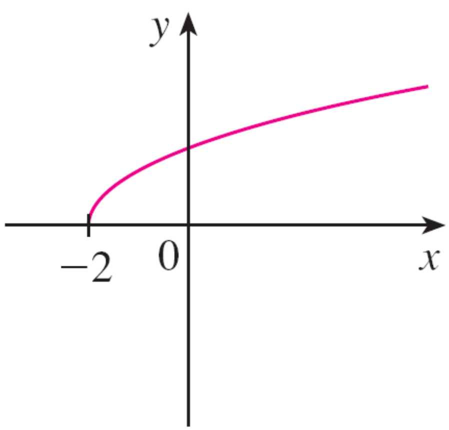
Figure 14(a)

# Representations of Functions

Notice that the equation $x = y^2 - 2$ implies $y^2 = x + 2$, so

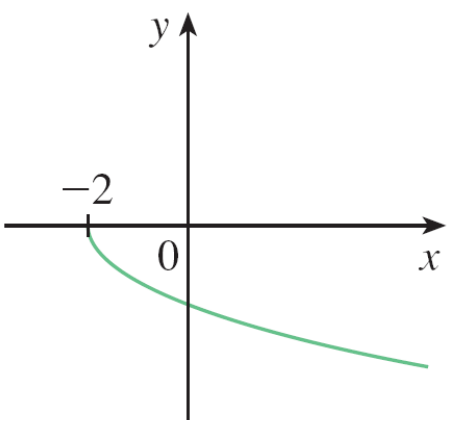
$$y = \pm\sqrt{x + 2}$$

Thus the upper and lower halves of the parabola are the graphs of the functions $f(x) = \sqrt{x + 2}$ and $g(x) = -\sqrt{x + 2}$. [See Figures 14(b) and (c).]




$y = \sqrt{x+2}$

**Figure 14(b)**




$y = -\sqrt{x+2}$

**Figure 14(c)**

# Representations of Functions

We observe that if we reverse the roles of $x$ and $y$, then the equation $x = h(y) = y^2 - 2$ *does* define $x$ as a function of $y$ (with $y$ as the independent variable and $x$ as the dependent variable) and the parabola now appears as the graph of the function $h$.

# Piecewise Defined Functions

# Example 7

A function $f$ is defined by

$$f(x) = \begin{cases} 1 - x & \text{if } x \le -1 \\ x^2 & \text{if } x > -1 \end{cases}$$

Evaluate $f(-2)$, $f(-1)$, and $f(0)$ and sketch the graph.

Solution:

Remember that a function is a rule. For this particular function the rule is the following:

First look at the value of the input $x$. If it happens that $x \le -1$, then the value of $f(x)$ is $1 - x$.

# Example 7 – *Solution*

cont'd

On the other hand, if $x > -1$, then the value of $f(x)$ is $x^2$.

Since $-2 \le -1$, we have $f(-2) = 1 - (-2) = 3$.

Since $-1 \le -1$, we have $f(-1) = 1 - (-1) = 2$.

Since $0 > -1$, we have $f(0) = 0^2 = 0$.

How do we draw the graph of $f$? We observe that if $x \le -1$, then $f(x) = 1 - x$, so the part of the graph of $f$ that lies to the left of the vertical line $x = -1$ must coincide with the line $y = 1 - x$, which has slope $-1$ and $y$-intercept 1.

# Example 7 – *Solution*

cont'd

If $x > -1$, then $f(x) = x^2$, so the part of the graph of $f$ that lies to the right of the line $x = -1$ must coincide with the graph of $y = x^2$, which is a parabola. This enables us to sketch the graph in Figure 15.

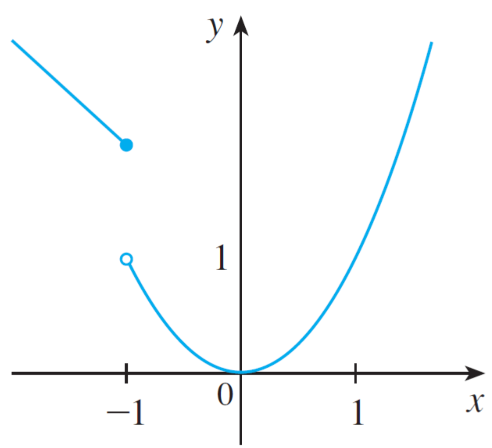



Figure 15

The solid dot indicates that the point $(-1, 2)$ is included on the graph; the open dot indicates that the point $(-1, 1)$ is excluded from the graph.

# Piecewise Defined Functions

The next example of a piecewise defined function is the absolute value function. Recall that the **absolute value** of a number *a*, denoted by $|a|$, is the distance from *a* to 0 on the real number line. Distances are always positive or 0, so we have

$$|a| \geq 0 \qquad \text{for every number } a$$

For example,

$|3| = 3$ $\quad |-3| = 3$ $\quad |0| = 0$ $\quad |\sqrt{2}\;1| = \sqrt{2}\;1$

$|3 - \pi| = \pi - 3$

# Piecewise Defined Functions

In general, we have

连续不可导

$$\boxed{\begin{aligned} |a| &= a && \text{if } a \geqslant 0 \\ |a| &= -a && \text{if } a < 0 \end{aligned}}$$

(Remember that if *a* is negative, then $-a$ is positive.)

# Example 8

Sketch the graph of the absolute value function $f(x) = |x|$.

Solution:

From the preceding discussion we know that

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

# Example 8 – *Solution*

cont'd

Using the same method as in Example 7, we see that the graph of $f$ coincides with the line $y = x$ to the right of the $y$-axis and coincides with the line $y = -x$ to the left of the $y$-axis (see Figure 16).

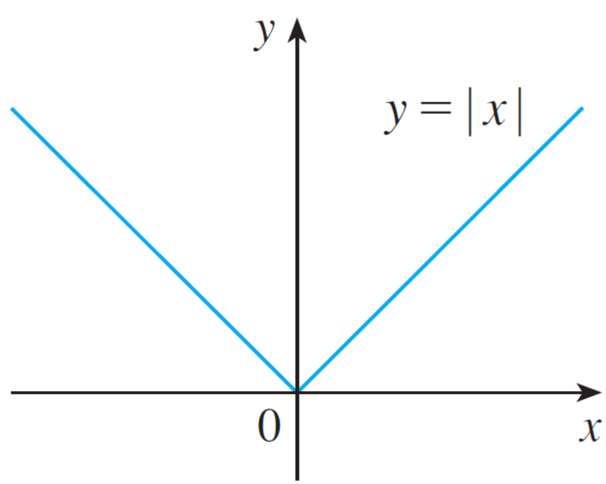



**Figure 16**

# Symmetry (odd/even function)

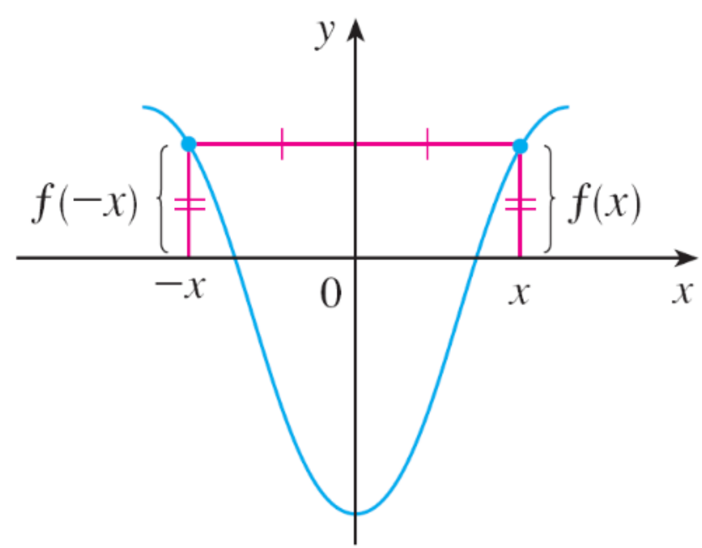
# Symmetry

If a function $f$ satisfies $f(-x) = f(x)$ for every number $x$ in its domain, then $f$ is called an **even function**.

For instance, the function $f(x) = x^2$ is even because

$$f(-x) = (-x)^2 = x^2 = f(x)$$

The geometric significance of an even function is that its graph is symmetric with respect to the $y$-axis (see Figure 19).




An even function

**Figure 19**

# Symmetry

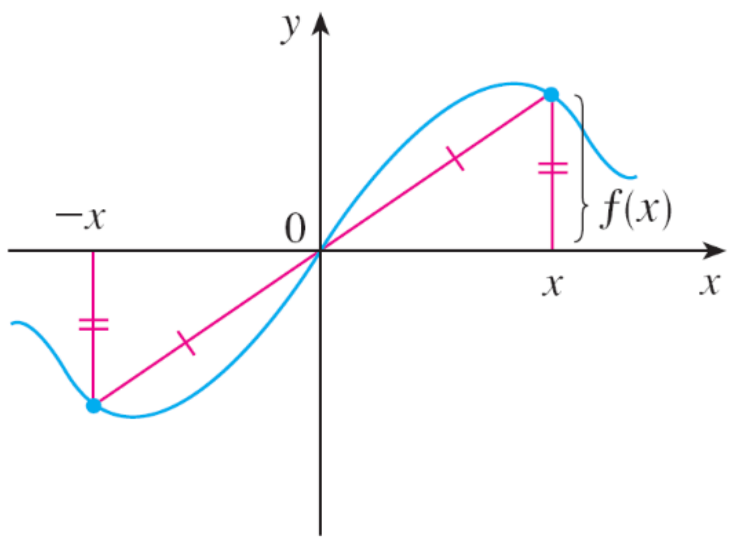
This means that if we have plotted the graph of $f$ for $x \geq 0$, we obtain the entire graph simply by reflecting this portion about the $y$-axis.

If $f$ satisfies $f(-x) = -f(x)$ for every number $x$ in its domain, then $f$ is called an **odd function**. For example, the function $f(x) = x^3$ is odd because

$$f(-x) = (-x)^3 = -x^3 = -f(x)$$

# Symmetry

The graph of an odd function is symmetric about the origin (see Figure 20).




An odd function

**Figure 20**

If we already have the graph of $f$ for $x \geq 0$, we can obtain the entire graph by rotating this portion through 180° about the origin.

# Example 11

Determine whether each of the following functions is even, odd, or neither even nor odd.

**(a)** $f(x) = x^5 + x$ odd **(b)** $g(x) = 1 - x^4$ even **(c)** $h(x) = 2x - x^2$

neither even nor odd

Solution:

**(a)** $f(-x) = (-x)^5 + (-x) = (-1)^5x^5 + (-x)$

$= -x^5 - x = -(x^5 + x)$

$= -f(x)$

Therefore $f$ is an odd function.

# Example 11 – *Solution*

cont'd

**(b)** $g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x)$

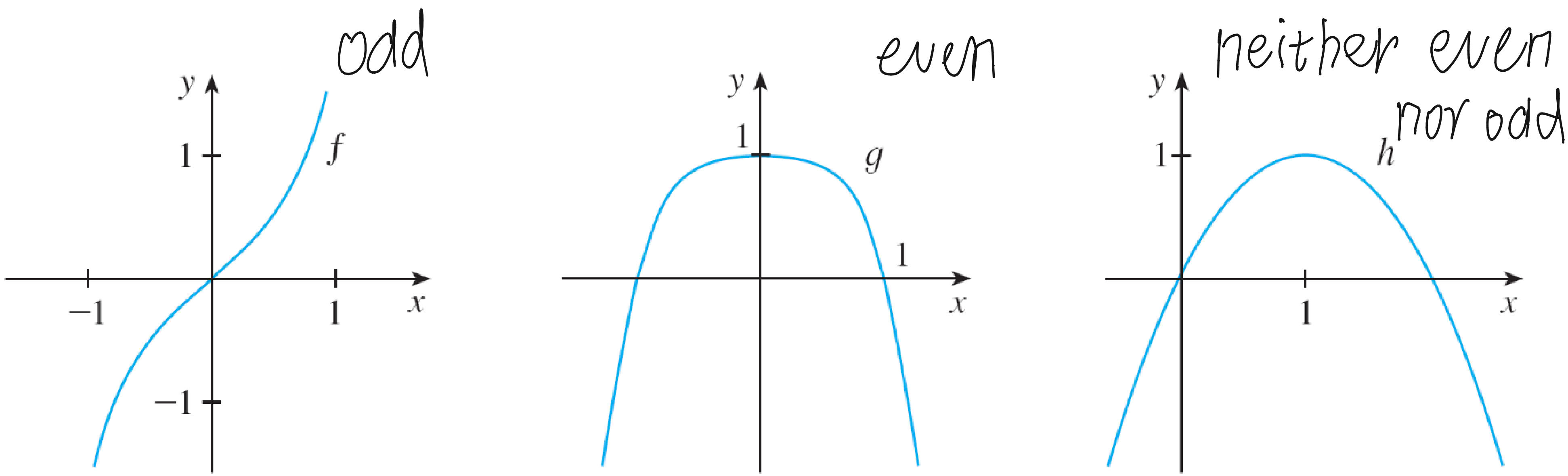
So $g$ is even.

**(c)** $h(-x) = 2(-x) - (-x)^2 = -2x - x^2$

Since $h(-x) \neq h(x)$ and $h(-x) \neq -h(x)$, we conclude that $h$ is neither even nor odd.

# Symmetry

The graphs of the functions in Example 11 are shown in Figure 21. Notice that the graph of *h* is symmetric neither about the *y*-axis nor about the origin.




(a) (b) (c)

**Figure 21**

# 1.2 Mathematical Models: A Catalog of Essential Functions

# Linear Models

When we say that $y$ is a **linear function** of $x$, we mean that the graph of the function is a line, so we can use the slope-intercept form of the equation of a line to write a formula for the function as

$$y = f(x) = mx + b$$

where $m$ is the slope of the line and $b$ is the $y$-intercept.

# Polynomials

A function $P$ is called a **polynomial** if

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where $n$ is a nonnegative integer and the numbers $a_0, a_1, a_2, \ldots, a_n$ are constants called the **coefficients** of the polynomial.

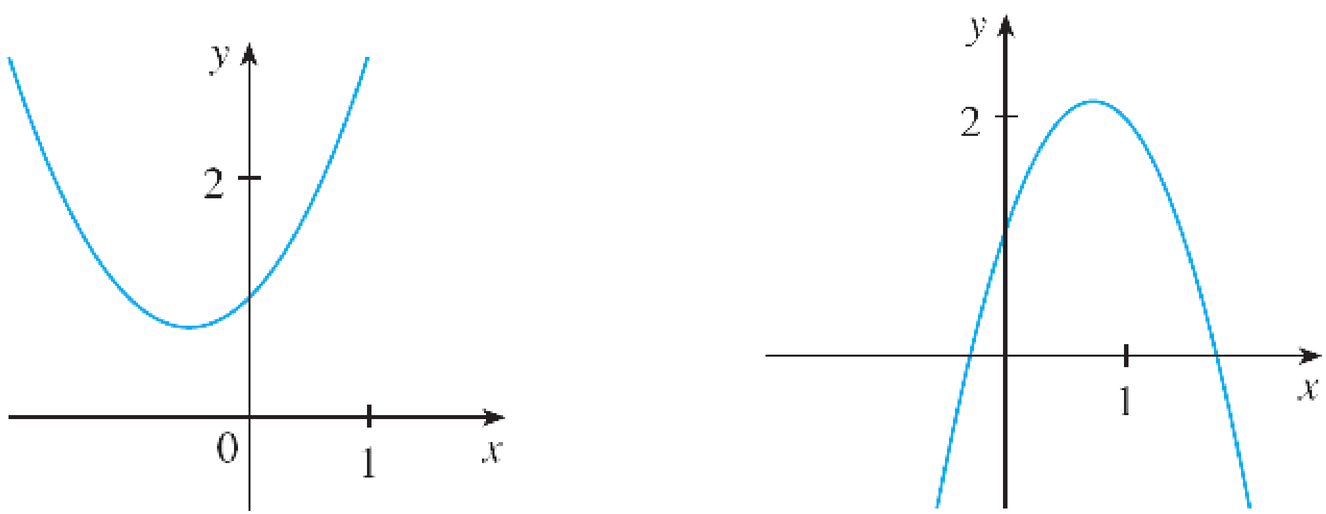
The domain of any polynomial is $\mathbb{R} = (-\infty, \infty)$. If the leading coefficient $a_n \neq 0$, then the **degree** [illegible]al is $n$. For example, the function

$$P(x) = 2x^6 - x^4 + \tfrac{2}{5}x^3 + \sqrt{2}$$

is a polynomial of degree 6.

# Polynomials

For a polynomial of degree 2, its graph is always a parabola obtained by shifting the parabola $y = ax^2$. The parabola opens upward if $a > 0$ and downward if $a < 0$. (See Figure 7.)




(a) $y = x^2 + x + 1$ (b) $y = -2x^2 + 3x + 1$

The graphs of quadratic functions are parabolas.

Figure 7

# Power Functions

A function of the form $f(x) = x^a$, where $a$ is a constant, is called a **power function**. We consider several cases.

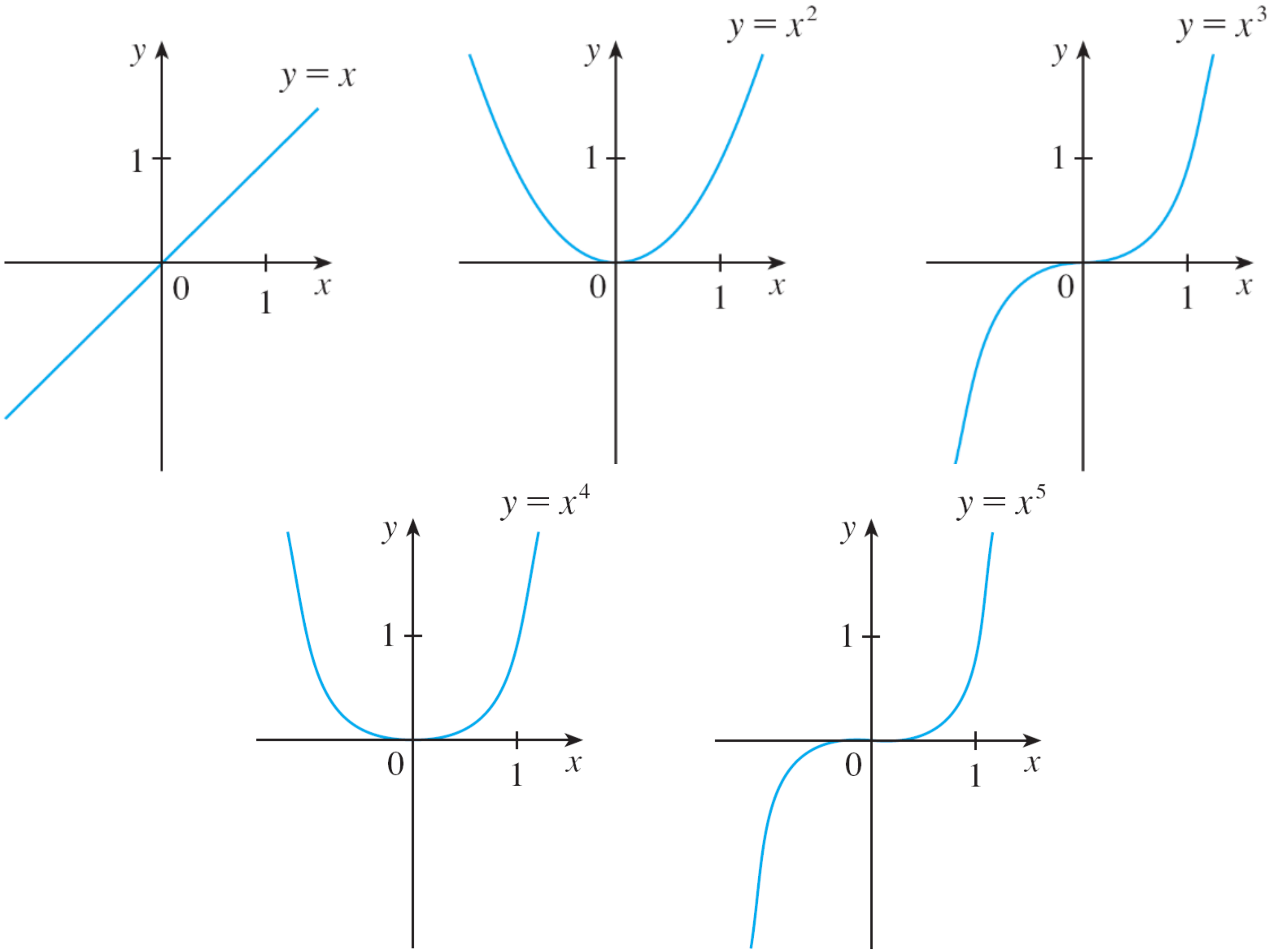
幂函数

**(i) $a = n$, where $n$ is a positive integer**

The graphs of $f(x) = x^n$ for $n = 1, 2, 3, 4$, and $5$ are shown in Figure 11. (These are polynomials with only one term.)

We already know the shape of the graphs of $y = x$ (a line through the origin with slope 1) and $y = x^2$ (a parabola).

# Power Functions



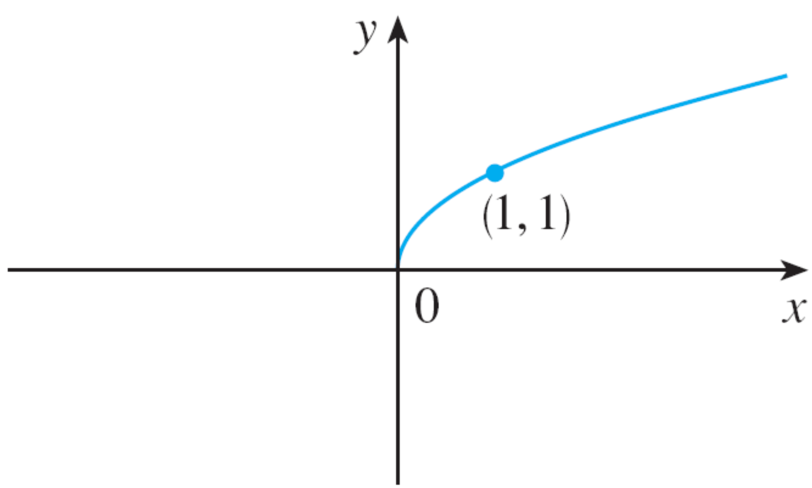

Graphs of $f(x) = x^n$ for $n = 1, 2, 3, 4, 5$

**Figure 11**

# Power Functions

## (ii) $a = 1/n$, where $n$ is a positive integer

The function $f(x) = x^{1/n} = \sqrt[n]{x}$ is a **root function**. For $n = 2$ it is the square root function $f(x) = \sqrt{x}$, whose domain is $[0, \infty)$ and whose graph is the upper half of the parabola $x = y^2$. [See Figure 13(a).]




$f(x) = \sqrt{x}$

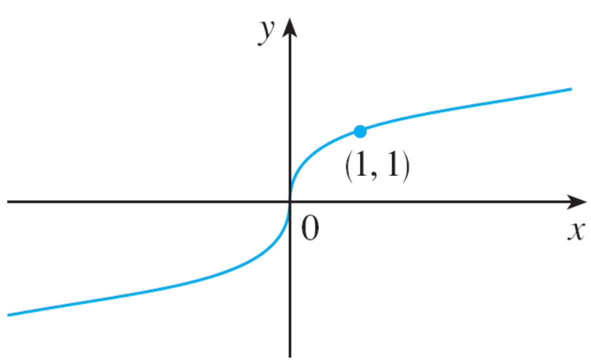
Graph of root function

**Figure 13(a)**

# Power Functions

For other even values of *n*, the graph of $y = \sqrt[n]{x}$ s similar to that of $y = \sqrt{x}$.

For *n* = 3 we have the cube root function $f(x) = \sqrt[3]{x}$ vhose domain is $\mathbb{R}$ (recall that every real number has a cube root) and whose graph is shown in Figure 13(b). The graph of $y = \sqrt[n]{x}$ 'or *n* odd (*n* > 3) is similar to that of $y = \sqrt[3]{x}$.




$f(x) = \sqrt[3]{x}$

Graph of root function

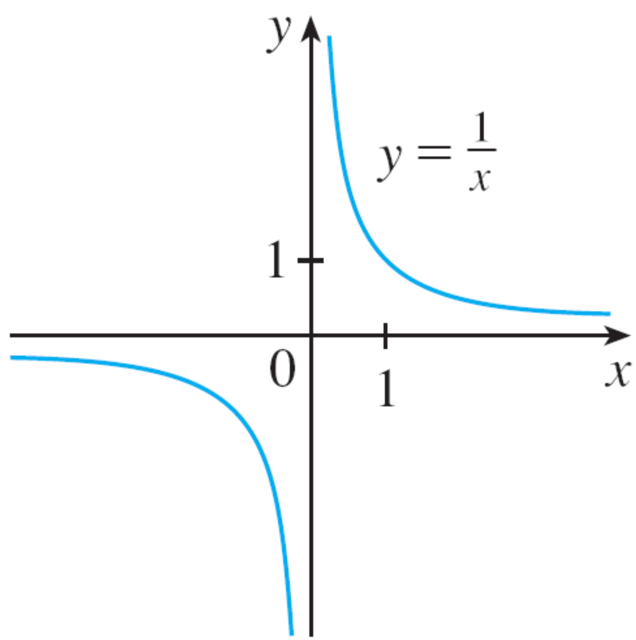
**Figure 13(b)**

# Power Functions

**(iii) $a = -1$**



The graph of the **reciprocal function** $f(x) = x^{-1} = 1/x$ is shown in Figure 14. Its graph has the equation $y = 1/x$, or $xy = 1$, and is a hyperbola with the coordinate axes as its asymptotes.




The reciprocal function

**Figure 14**

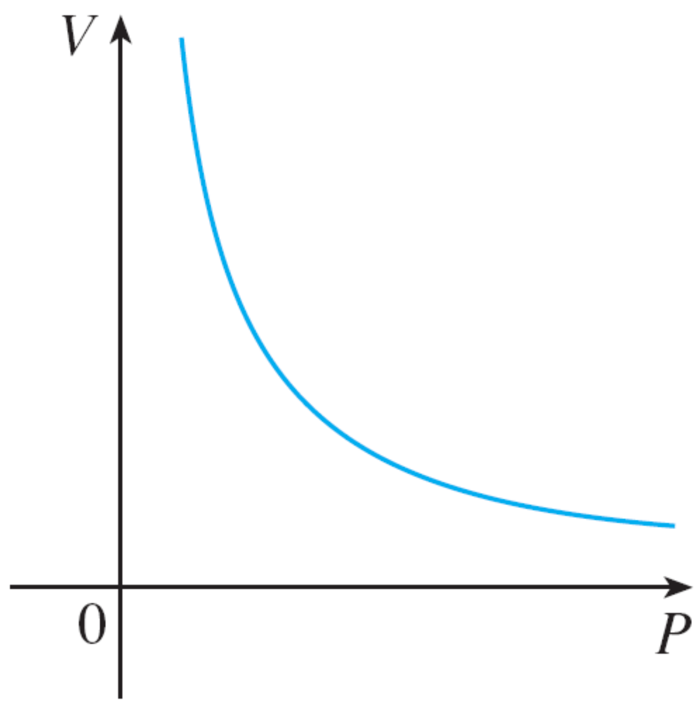
# Power Functions

This function arises in physics and chemistry in connection with Boyle's Law, which says that, when the temperature is constant, the volume $V$ of a gas is inversely proportional to the pressure $P$:

$$V = \frac{C}{P}$$

where $C$ is a constant.

Thus the graph of $V$ as a function of $P$ (see Figure 15) has the same general shape as the right half of Figure 14.




Volume as a function of pressure at constant temperature

**Figure 15**

# Rational Functions

A **rational function** *f* is a ratio of two polynomials:

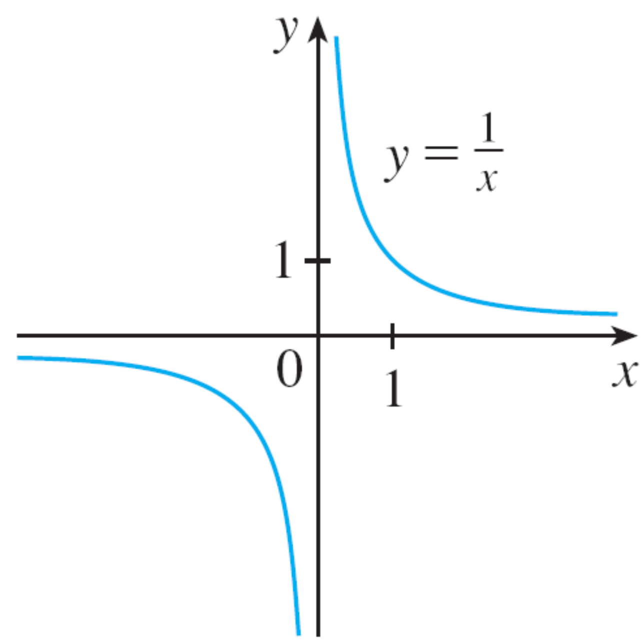
有理函数

$$f(x) = \frac{P(x)}{Q(x)}$$

$f(x) = \frac{x^2+\sqrt{x}}{x+x^3}$ 非有理函数

where *P* and *Q* are polynomials.
The domain consists of all values of $x$ such that $Q(x) \neq 0$.

A simple example of a rational function is the function $f(x) = 1/x$, whose domain is $\{x \mid x \neq 0\}$; this is the reciprocal function graphed in Figure 14.




The reciprocal function

Figure 14

# Rational Functions

The function

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$

有理函数

is a rational function with domain $\{x \mid x \neq \pm 2\}$. Its graph is shown in Figure 16.

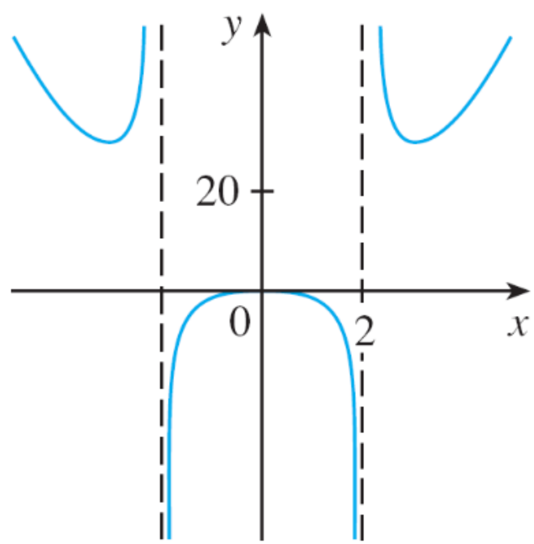



Figure 16

# Trigonometric Functions 三角函数

In calculus the convention is that radian measure is always used (except when otherwise indicated).

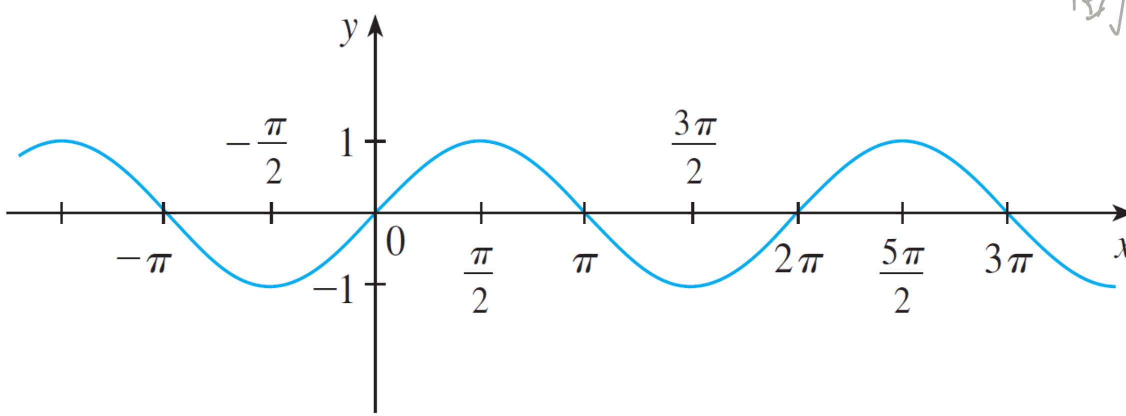
For example, when we use the function $f(x) = \sin x$, it is understood that $\sin x$ means the sine of the angle whose radian measure is $x$.

角度是弧度,不是角度

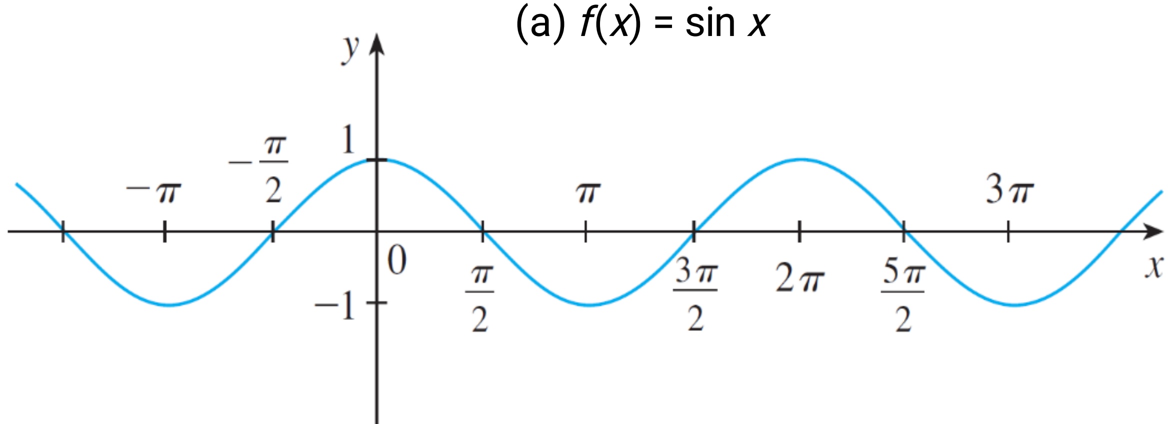
# Trigonometric Functions

Thus the graphs of the sine and cosine functions are as shown in Figure 18.

周期函数




(a) $f(x) = \sin x$




(b) $g(x) = \cos x$

**Figure 18**

# Trigonometric Functions

Notice that for both the sine and cosine functions the domain is $(-\infty, \infty)$ and the range is the closed interval $[-1, 1]$.

Thus, for all values of $x$, we have

$$\boxed{-1 \leqslant \sin x \leqslant 1 \qquad -1 \leqslant \cos x \leqslant 1}$$

or, in terms of absolute values,

$$|\sin x| \leq 1 \qquad |\cos x| \leq 1$$

# Trigonometric Functions

Also, the zeros of the sine function occur at the integer multiples of $\pi$; that is,

$$\sin x = 0 \qquad \text{when} \qquad x = n\pi \qquad n \text{ an integer}$$

An important property of the sine and cosine functions is that they are periodic functions and have period $2\pi$.

This means that, for all values of $x$,

$$\boxed{\sin(x + 2\pi) = \sin x \qquad \cos(x + 2\pi) = \cos x}$$

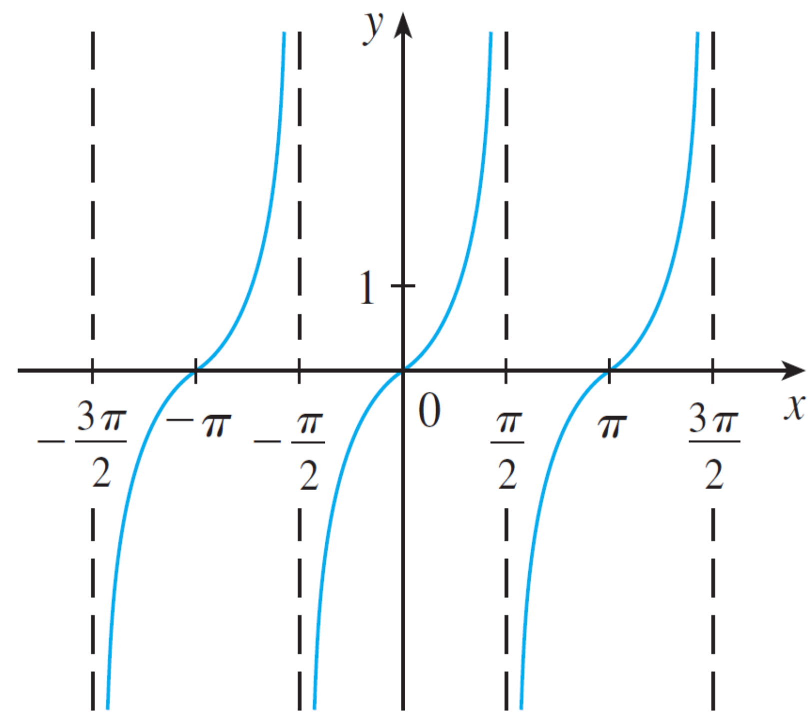
# Trigonometric Functions

The tangent function is related to the sine and cosine functions by the equation

$$\tan x = \frac{\sin x}{\cos x}$$

and its graph is shown in Figure 19. It is undefined whenever $\cos x = 0$, that is, when $x = \pm\pi/2, \pm 3\pi/2, \ldots$.

Its range is $(-\infty, \infty)$




$y = \tan x$

**Figure 19**

# Trigonometric Functions

Notice that the tangent function has period $\pi$:

$$\tan(x + \pi) = \tan x \qquad \text{for all } x$$

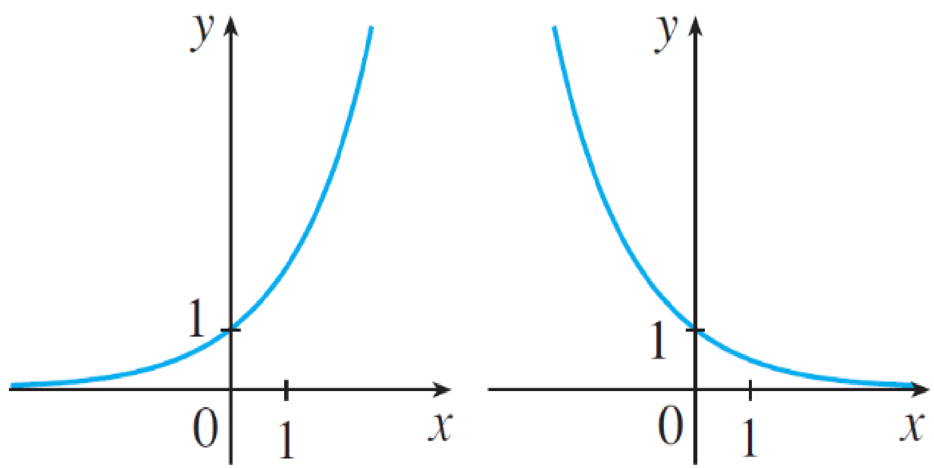
The remaining three trigonometric functions (cosecant, secant, and cotangent) are the reciprocals of the sine, cosine, and tangent functions.

# Exponential Functions

指数函数

The **exponential functions** are the functions of the form $f(x) = b^x$, where the base $b$ is a positive constant.

The graphs of $y = 2^x$ and $y = (0.5)^x$ are shown in Figure 20. In both cases the domain is $(-\infty, \infty)$ and the range is $(0, \infty)$.




(a) $y = 2^x$ (b) $y = (0.5)^x$

**Figure 20**

# Exponential Functions

Exponential functions are useful for modeling many natural phenomena, such as population growth (if $b > 1$) and radioactive decay (if $b < 1$). 单调递增

单调递减

# Logarithmic Functions

对数函数

The **logarithmic functions** $f(x) = \log_b x$, where the base $b$ is a positive constant, are the inverse functions of the exponential functions. Figure 21 shows the graphs of four logarithmic functions with various bases.

In each case the domain is $(0, \infty)$, the range is $(-\infty, \infty)$, and the function increases slowly when $x > 1$.

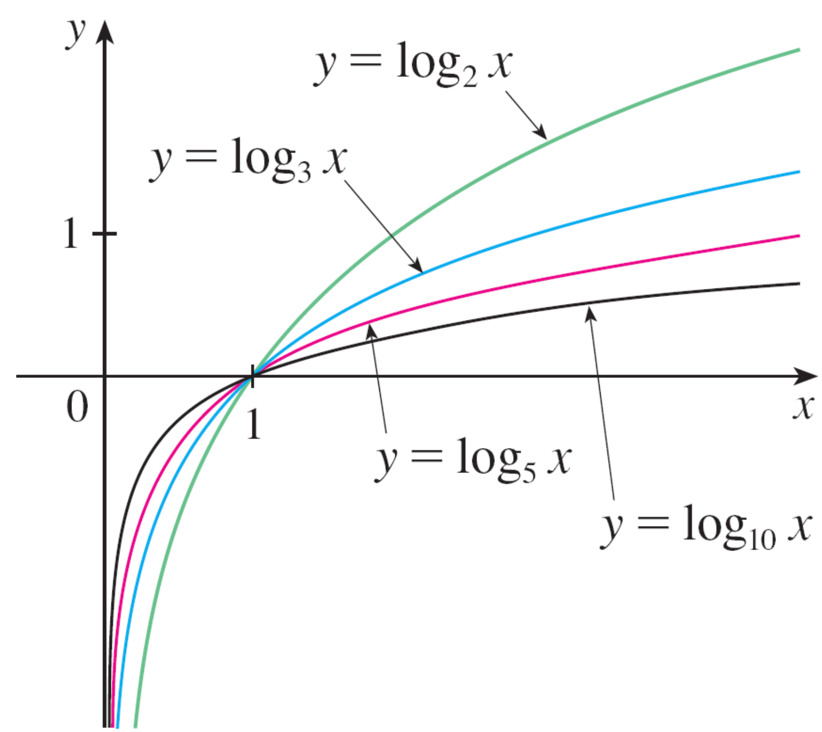



Figure 21

# Example 6

Classify the following functions as one of the types of functions that we have discussed.

**(a)** $f(x) = 5^x$

**(b)** $g(x) = x^5$

**(c)** $h(x) = \dfrac{1 + x}{1 - \sqrt{x}}$

**(d)** $u(t) = 1 - t + 5t^4$

# 1.3 New Functions from Old

# Transformations of Functions

**Vertical and Horizontal Shifts** Suppose $c > 0$. To obtain the graph of

垂直 水平

$y = f(x) + c$, shift the graph of $y = f(x)$ a distance $c$ units upward

$y = f(x) - c$, shift the graph of $y = f(x)$ a distance $c$ units downward

$y = f(x - c)$, shift the graph of $y = f(x)$ a distance $c$ units to the right

$y = f(x + c)$, shift the graph of $y = f(x)$ a distance $c$ units to the left

**Vertical and Horizontal Stretching and Reflecting** Suppose $c > 1$. To obtain the graph of

$y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor of $c$

$y = (1/c)f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of $c$

$y = f(cx)$, shrink the graph of $y = f(x)$ horizontally by a factor of $c$

$y = f(x/c)$, stretch the graph of $y = f(x)$ horizontally by a factor of $c$

$y = -f(x)$, reflect the graph of $y = f(x)$ about the $x$-axis

$y = f(-x)$, reflect the graph of $y = f(x)$ about the $y$-axis

# Transformations of Functions

Figure 3 illustrates these stretching transformations when applied to the cosine function with $c = 2$.

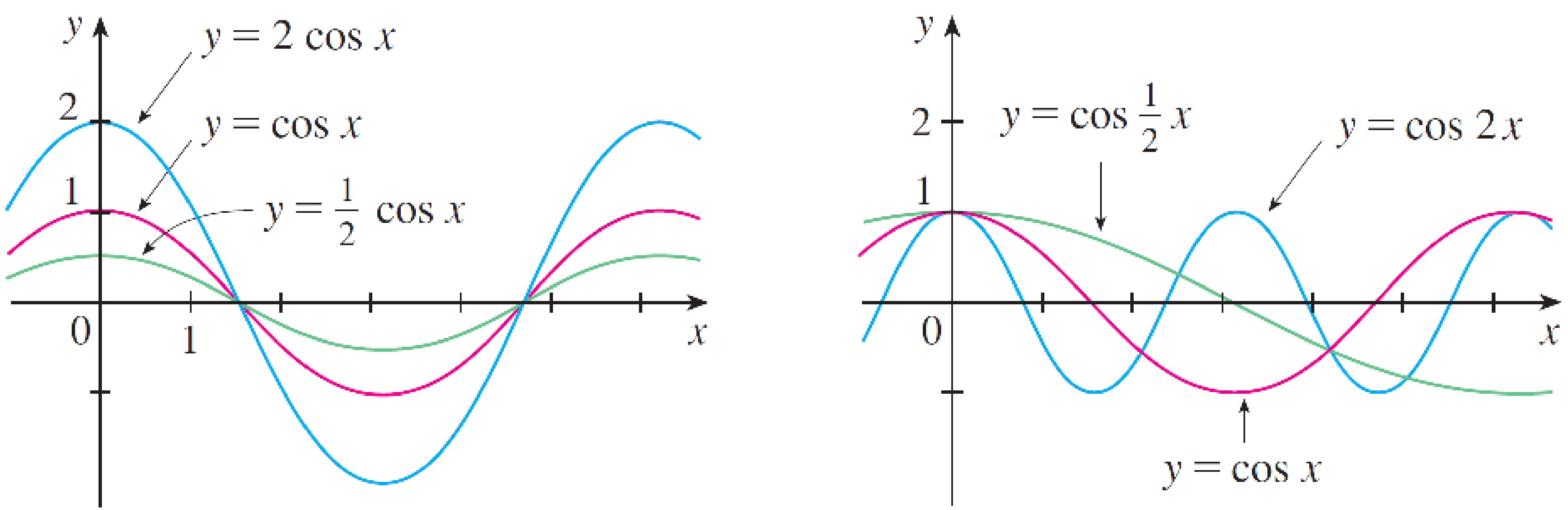



Figure 3

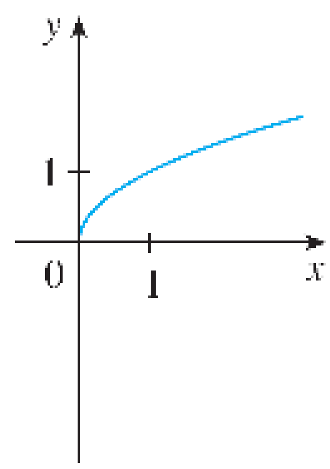
# Example 1

Given the graph of $y = \sqrt{x}$, use transformations to graph

$y = \sqrt{x} - 2$, $y = \sqrt{x - 2}$, $y = -\sqrt{x}$, $y = 2\sqrt{x}$, $y = \sqrt{-x}$.

Solution:

The graph of the square root function $y = \sqrt{x}$ is shown in Figure 4(a).




(a) $y = \sqrt{x}$

Figure 4

# Example 1 – *Solution*

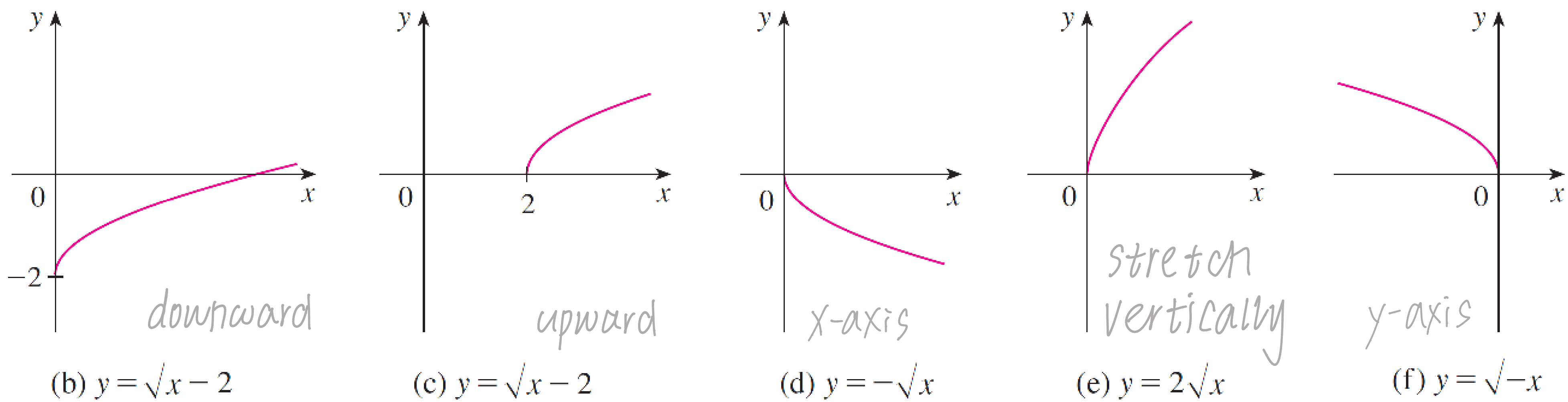



Figure 4

In the other parts of the figure we sketch $y = \sqrt{x} - 2$ by shifting 2 units downward, $y = \sqrt{x - 2}$ by shifting 2 units to the right, $y = -\sqrt{x}$ by reflecting about the *x*-axis, $y = 2\sqrt{x}$ by stretching vertically by a factor of 2, and $y = \sqrt{-x}$ by reflecting about the *y*-axis.

# Transformations of Functions

Another transformation of some interest is taking the *absolute value* of a function. If $y = |f(x)|$, then according to the definition of absolute value, $y = f(x)$ when $f(x) \geq 0$ and $y = -f(x)$ when $f(x) < 0$.

This tells us how to get the graph of $y = |f(x)|$ from the graph of $y = f(x)$: The part of the graph that lies above the $x$-axis remains the same; the part that lies below the $x$-axis is reflected about the $x$-axis.

# Combinations of Functions

# Combinations of Functions

Two functions $f$ and $g$ can be combined to form new functions $f + g$, $f - g$, $fg$, and $f/g$ in a manner similar to the way we add, subtract, multiply, and divide real numbers. The sum and difference functions are defined by

$$(f + g)(x) = f(x) + g(x) \qquad (f - g)(x) = f(x) - g(x)$$

If the domain of $f$ is $A$ and the domain of $g$ is $B$, then the domain of $f + g$ is the intersection $A \cap B$ because both $f(x)$ and $g(x)$ have to be defined.

或$f-g$或$f\cdot g$或$\frac{f}{g}$

都为A与B的交集

For example, the domain of $f(x) = \sqrt{x}$ is $A = [0, \infty)$ and the domain of $g(x) = \sqrt{2 - x}$ is $B = (-\infty, 2]$, so the domain of $(f + g)(x) = \sqrt{x} + \sqrt{2 - x}$ is $A \cap B = [0, 2]$.

# Combinations of Functions

Similarly, the product and quotient functions are defined by

$$(fg)(x) = f(x)g(x) \qquad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$ 商函数

The domain of $fg$ is $A \cap B$, but we can't divide by 0 and so the domain of $f/g$ is $\{x \in A \cap B \mid g(x) \neq 0\}$.

For instance, if $f(x) = x^2$ and $g(x) = x - 1$, then the domain of the rational function $(f/g)(x) = x^2/(x - 1)$ is $\{x \mid x \neq 1\}$, or $(-\infty, 1)$ U $(1, \infty$

# Combinations of Functions 复合函数

There is another way of combining two functions to obtain a new function. For example, suppose that $y = f(u) = \sqrt{u}$ and $u = g(x) = x^2 + 1$.

Since $y$ is a function of $u$ and $u$ is, in turn, a function of $x$, it follows that $y$ is ultimately a function of $x$. We compute this by substitution:

$$y = f(u) = f(g(x)) = f(x^2 + 1) = \sqrt{x^2 + 1}$$

g first f second

The procedure is called *composition* because the new function is *composed* of the two given functions $f$ and $g$.

# Combinations of Functions

In general, given any two functions $f$ and $g$, we start with a number $x$ in the domain of $g$ and calculate $g(x)$. If this number $g(x)$ is in the domain of $f$, then we can calculate the value of $f(g(x))$.

The result is a new function $h(x) = f(g(x))$ obtained by substituting $g$ into $f$. It is called the *composition*

(or *composite*) of $f$ and $g$ and is denoted by $f \circ g$ ("$f$ circle $g$").

**Definition** Given two functions $f$ and $g$, the **composite function** $f \circ g$ (also called the **composition** of $f$ and $g$) is defined by

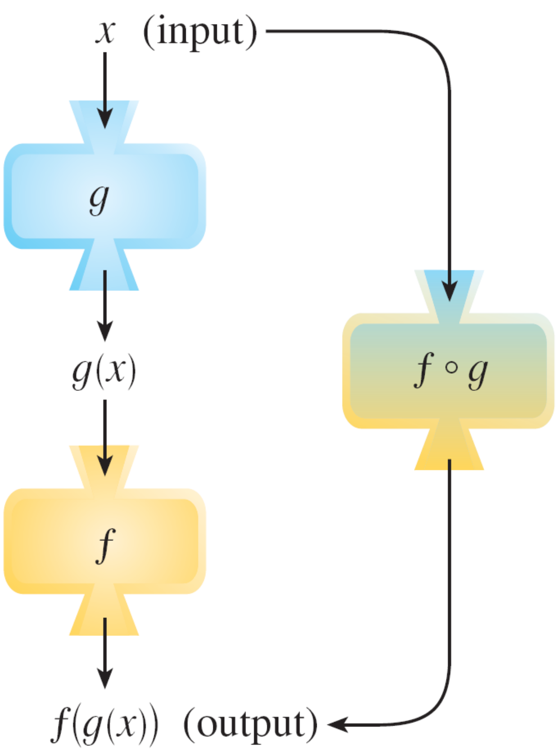
$$(f \circ g)(x) = f(g(x))$$

# Combinations of Functions

The domain of $f \circ g$ is the set of all $x$ in the domain of $g$ such that $g(x)$ is in the domain of $f$.

In other words, $(f \circ g)(x)$ is defined whenever both $g(x)$ and $f(g(x))$ are defined.

Figure 11 shows how to picture $f \circ g$ in terms of machines.




The $f \circ g$ machine is composed of the $g$ machine (first) and then the $f$ machine.

**Figure 11**

# Example 6

If $f(x) = x^2$ and $g(x) = x - 3$, find the composite functions $f \circ g$ and $g \circ f$.

Solution:

We have

$$(f \circ g)(x) = f(g(x)) = f(x - 3) = (x - 3)^2$$

$$= g(x^2) = x^2 - 3$$

$$(g \circ f)(x) = g(f(x))$$

# Combinations of Functions

We know that, the notation $f \circ g$ means that the function $g$ is
applied first and then $f$ is applied second. In Example 6,

$f \circ g$ is the function that *first* subtracts 3 and *then* squares;

$g \circ f$ is the function that *first* squares and *then* subtracts 3.

It is possible to take the composition of three or more functions. For instance, the composite function $f \circ g \circ h$ is found by first applying $h$, then $g$, and then $f$ as follows:

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

# 1.4 Exponential Functions

# Exponential Functions

The function $f(x) = 2^x$ is called an *exponential function* because the variable, $x$, is the exponent. It should not be confused with the power function $g(x) = x^2$, in which the variable is the base.

In general, an **exponential function** is a function of the form

$$f(x) = b^x$$

where $b$ is a positive constant. Let's recall what this means. If $x = n$, a positive integer, then

$$b^n = \underbrace{b \cdot b \cdot \cdots \cdot b}_{n \text{ factors}}$$

# Exponential Functions

If $x = 0$, then $b^0 = 1$, and if $x = -n$, where $n$ is a positive integer, then

$$b^{-n} = \frac{1}{b^n}$$



If $x$ is a rational number, $x = p/q$, where $p$ and $q$ are integers and $q > 0$, then

$$b^x = b^{p/q} = \sqrt[q]{b^p} = \left(\sqrt[q]{b}\right)^p$$

# Exponential Functions

The graphs of members of the family of functions $y = b^x$ are shown in Figure 3 for various values of the base $b$.

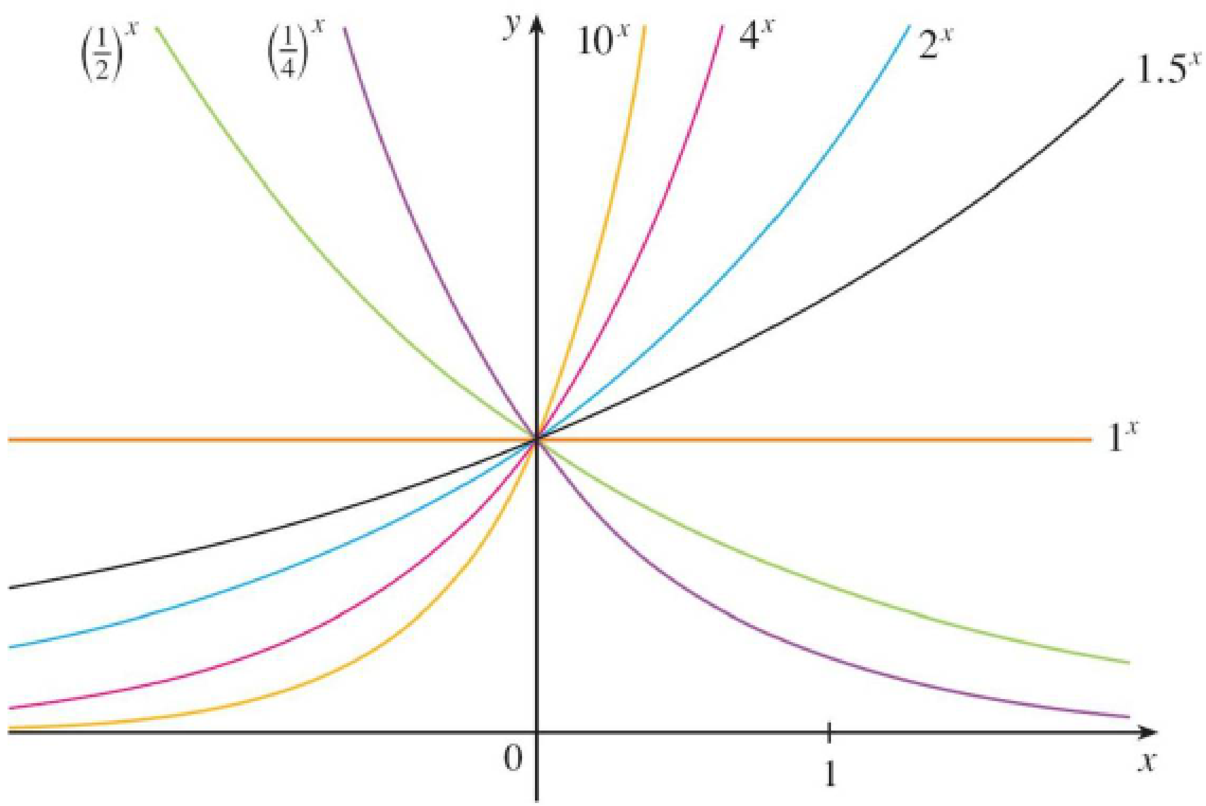



Figure 3

# Exponential Functions

Observe that if $b \neq 1$, then the exponential function $y = b^x$ has domain $\mathbb{R}$ and range $(0, \infty)$.

Notice also that, since $(1/b)^x = 1/b^x = b^{-x}$, the graph of $y = (1/b)^x$ is just the reflection of the graph of $y = b^x$ about the $y$-axis.

# Exponential Functions

One reason for the importance of the exponential function lies in the following properties.

If *x* and *y* are rational numbers, then these laws are well known from elementary algebra. It can be proved that they remain true for arbitrary real numbers *x* and *y*.

**Laws of Exponents** If $a$ and $b$ are positive numbers and $x$ and $y$ are any real numbers, then

**1.** $b^{x+y} = b^x b^y$ **2.** $b^{x-y} = \dfrac{b^x}{b^y}$ **3.** $(b^x)^y = b^{xy}$ **4.** $(ab)^x = a^x b^x$

# The Number *e*

Of all possible bases for an exponential function, there is one that is most convenient for the purposes of calculus.

The choice of a base $b$ is influenced by the way the graph of $y = b^x$ crosses the $y$-axis.

# The Number $e$

Figures 13 and 14 show the tangent lines to the graphs of $y = 2^x$ and $y = 3^x$ at the point (0, 1).

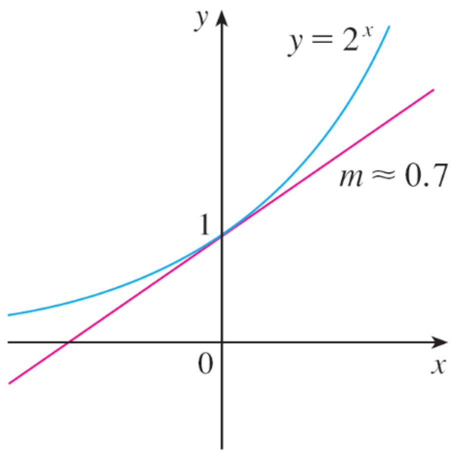



Figure 13

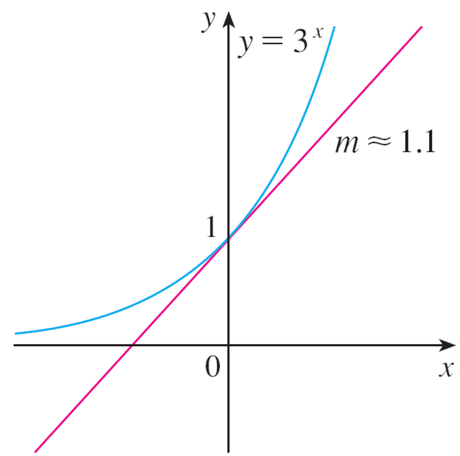
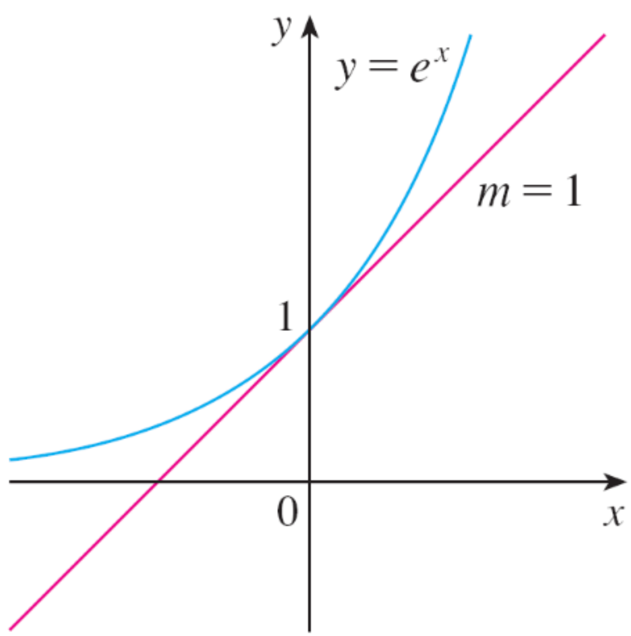



Figure 14

(For present purposes, you can think of the tangent line to an exponential graph at a point as the line that touches the graph only at that point.) If we measure the slopes of these tangent lines at (0, 1), we find that $m \approx 0.7$ for $y = 2^x$ and $m \approx 1.1$ for $y = 3^x$.

# The Number $e$

It turns out, that some of the formulas of calculus will be greatly simplified if we choose the base $b$ so that the slope of the tangent line to $y = b^x$ at (0, 1) is *exactly* 1.
(See Figure 15.)




The natural exponential function crosses the $y$-axis with a slope of 1.

**Figure 15**

# The Number $e$

In fact, there *is* such a number and it is denoted by the letter $e$. (This notation was chosen by the Swiss mathematician Leonhard Euler in 1727, probably because it is the first letter of the word *exponential*.)

# The Number $e$

In view of Figures 13 and 14, it comes as no surprise that the number $e$ lies between 2 and 3 and the graph of $y = e^x$ lies between the graphs of $y = 2^x$ and $y = 3^x$.
(See Figure 16.)

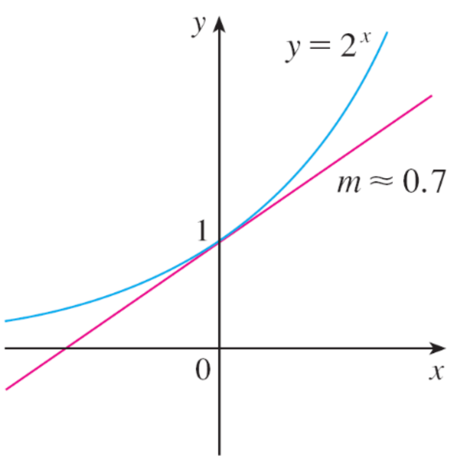



Figure 13

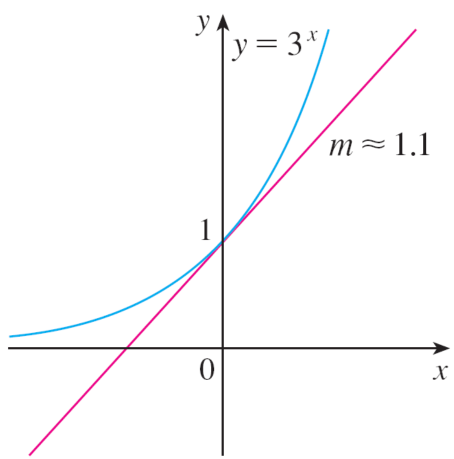



Figure 14

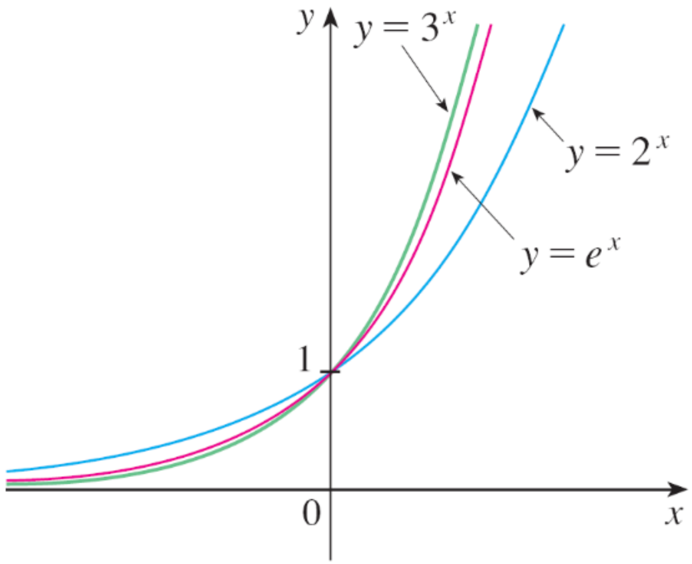



Figure 16

# The Number $e$

We will see that the value of $e$, correct to five decimal places, is

$$e \approx 2.71828$$

We call the function $f(x) = e^x$ the **natural exponential function**.

自然指数函数

# Example 3

The half-life of strontium-90, $^{90}Sr$, is 25 years. This means that half of any given quantity of $^{90}Sr$ will disintegrate in 25 years.

(a)If a sample of $^{90}Sr$ has a mass of 24 mg, find an expression for the mass $m(t)$ that remains after $t$ years.

(b)Find the mass remaining after 40 years, correct to the nearest milligram.

(c) Use a graphing device to graph $m(t)$ and use the graph to estimate the time required for the mass to be reduced to 5 mg.

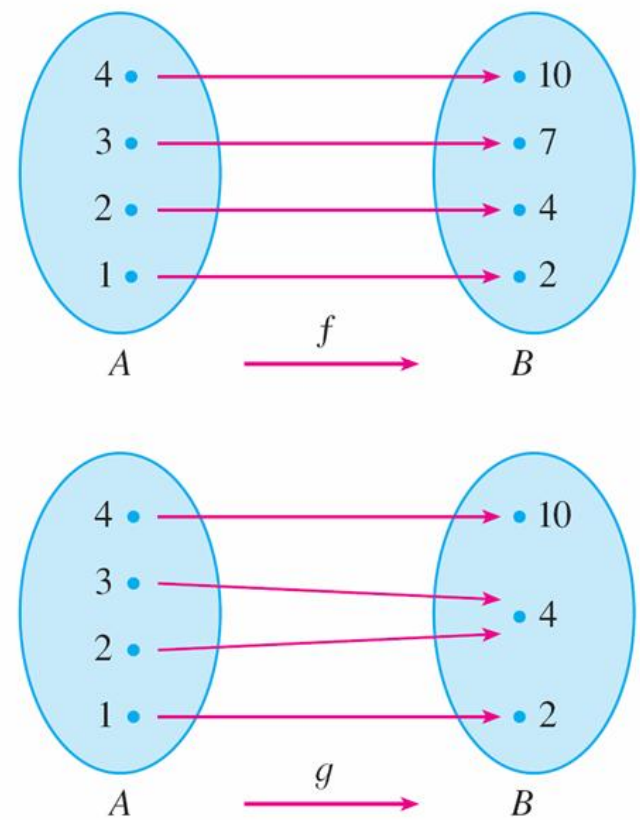
# 1.5 Inverse Functions and Logarithms

反函数

# Inverse Functions and Logarithms

Let's compare the functions $f$ and $g$ whose arrow diagrams are shown in Figure 1.

one to one




$f$ is one-to-one; $g$ is not.

**Figure 1**

Note that $f$ never takes on the same value twice (any two inputs in $A$ have different outputs), whereas $g$ does take on the same value twice (both 2 and 3 have the same output, 4).

In symbols,

but $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$

# Inverse Functions and Logarithms

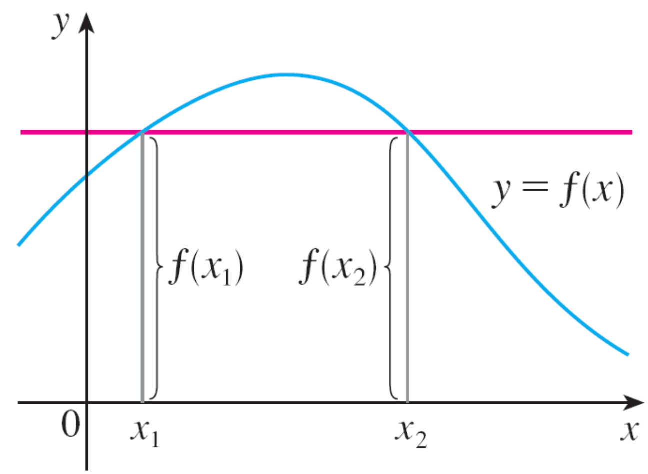
Functions that share this property with $f$ are called *one-to-one functions*.

**1 Definition** A function $f$ is called a **one-to-one function** if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \qquad \text{whenever } x_1 \neq x_2$$

# Inverse Functions and Logarithms

If a horizontal line intersects the graph of $f$ in more than one point, then we see from Figure 2 that there are numbers $x_1$ and $x_2$ such that $f(x_1) = f(x_2)$.
This means that $f$ is not one-to-one.




This function is not one-to-one because $f(x_1) = f(x_2)$.

**Figure 2**

Therefore we have the following geometric method for determining whether a function is one-to-one.

**Horizontal Line Test** A function is one-to-one if and only if no horizontal line intersects its graph more than once.

# Inverse Functions and Logarithms

One-to-one functions are important because they are precisely the functions that possess inverse functions according to the following definition.

**2 Definition** Let $f$ be a one-to-one function with domain $A$ and range $B$. Then its **inverse function** $f^{-1}$ has domain $B$ and range $A$ and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any $y$ in $B$.

This definition says that if $f$ maps $x$ into $y$, then $f^{-1}$ maps $y$ back into $x$. (If $f$ were not one-to-one, then $f^{-1}$ would not be uniquely defined.)

# Inverse Functions and Logarithms

The arrow diagram in Figure 5 indicates that $f^{-1}$ reverses the effect of $f$.

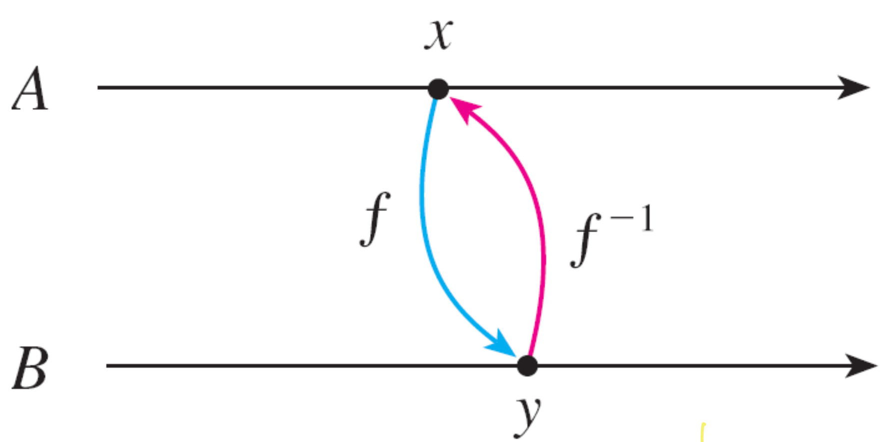



Figure 5

$\frac{1}{\sin(x)} = (\sin(x))^{-1} \neq \sin^{-1}(x)$

Note that

$$\text{domain of } f^{-1} = \text{range of } f$$

$$\text{range of } f^{-1} = \text{domain of } f$$

# Inverse Functions and Logarithms

For example, the inverse function of $f(x) = x^3$ is $f^{-1}(x) = x^{1/3}$ because if $y = x^3$, then

$$f^{-1}(y) = f^{-1}(x^3) = (x^3)^{1/3} = x$$

**Caution**

Do not mistake the $-1$ in $f^{-1}$ for an exponent. Thus

$$f^{-1}(x) \text{ does } \textit{not} \text{ mean } \frac{1}{f(x)}$$

The reciprocal $1/f(x)$ could, however, be written as $[f(x)]^{-1}$.

# Example

If $f(1) = 5$, $f(3) = 7$, and $f(8) = -10$, find $f^{-1}(7)$, $f^{-1}(5)$, and $f^{-1}(-10)$.

Solution:

From the definition of $f^{-1}$ we have

| | | |
|---|---|---|
| $f^{-1}(7) = 3$ | because | $f(3) = 7$ |
| $f^{-1}(5) = 1$ | because | $f(1) = 5$ |
| $f^{-1}(-10) = 8$ | because | $f(8) = -10$ |

# Inverse Functions and Logarithms

The letter $x$ is traditionally used as the independent variable, so when we concentrate on $f^{-1}$ rather than on $f$, we usually reverse the roles of $x$ and $y$ in Definition 2 and write

**3** $$f^{-1}(x) = y \iff f(y) = x$$

By substituting for $y$ in Definition 2 and substituting for $x$ in (3), we get the following **cancellation equations**:

**4** $$\begin{aligned} f^{-1}(f(x)) &= x \quad \text{for every } x \text{ in } A \\ f(f^{-1}(x)) &= x \quad \text{for every } x \text{ in } B \end{aligned}$$

# Inverse Functions and Logarithms

$x \to \boxed{f} \xrightarrow{y} \boxed{f^{-1}} \xrightarrow{x} \boxed{f} \xrightarrow{y}$

The graph of $f^{-1}$ is obtained by reflecting the graph of $f$ about the line $y = x$.

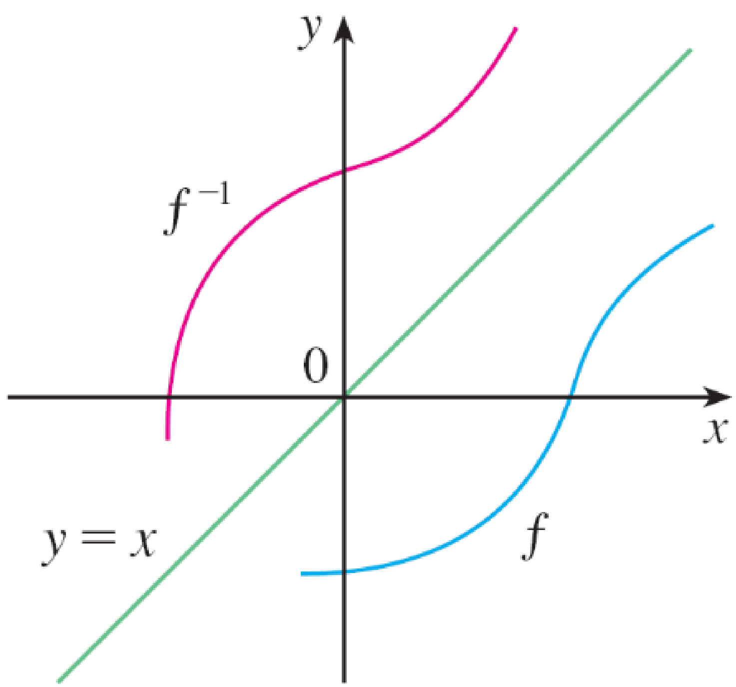



Figure 9

# Logarithmic Functions

If $b > 0$ and $b \neq 1$, the exponential function $f(x) = b^x$ is either increasing or decreasing and so it is one-to-one by the Horizontal Line Test. It therefore has an inverse function $f^{-1}$, which is called the **logarithmic function with base *b*** and is denoted by $\log_b$.

If we use the formulation of an inverse function given by $\boxed{3}$,

$$f^{-1}(x) = y \iff f(y) = x$$

then

$\boxed{6}$

$$\boxed{\log_b x = y \iff b^y = x}$$

# Logarithmic Functions

Thus, if $x > 0$, then $\log_b x$ is the exponent to which the base $b$ must be raised to give $x$.

For example, $\log_{10} 0.001 = -3$ because $10^{-3} = 0.001$.

The cancellation equations (4), when applied to the functions $f(x) = b^x$ and $f^{-1}(x) = \log_b x$, become

**7**

$$\log_b(b^x) = x \quad \text{for every } x \in \mathbb{R}$$
$$b^{\log_b x} = x \quad \text{for every } x > 0$$

# Logarithmic Functions

The logarithmic function $\log_b$ has domain $(0, \infty)$ and range $\mathbb{R}$. Its graph is the reflection of the graph of $y = b^x$ about the line $y = x$.

Figure 11 shows the case where $b > 1$. (The most important logarithmic functions have base $b > 1$.)

The fact that $y = b^x$ is a very rapidly increasing function for $x > 0$ is reflected in the fact that $y = \log_b x$ is a very slowly increasing function for $x > 1$.

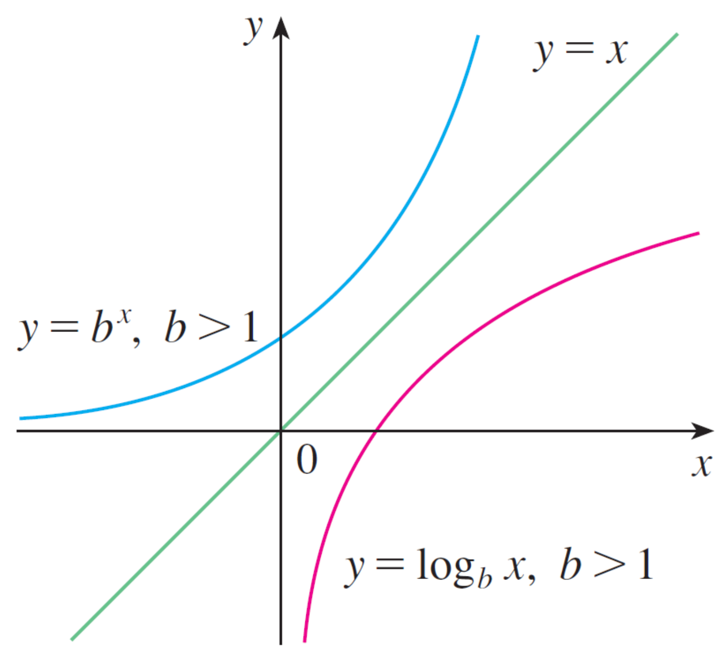



Figure 11

# Logarithmic Functions

Figure 12 shows the graphs of $y = \log_b x$ with various values of the base $b > 1$.

Since $\log_b 1 = 0$, the graphs of all logarithmic functions pass through the point (1, 0).

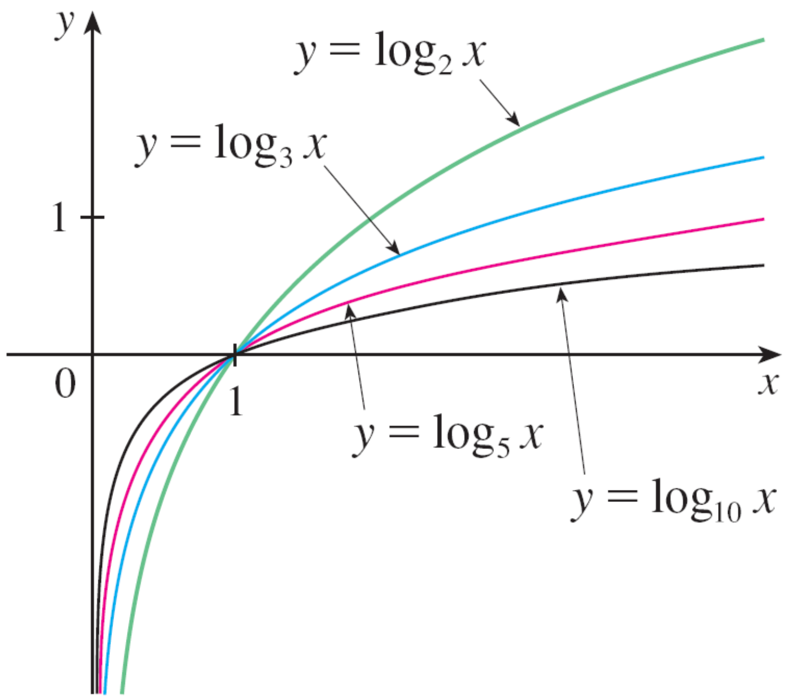



Figure 12

# Logarithmic Functions

The following properties of logarithmic functions follow from the corresponding properties of exponential functions.

**Laws of Logarithms** If $x$ and $y$ are positive numbers, then

1. $\log_b(xy) = \log_b x + \log_b y$
2. $\log_b\left(\dfrac{x}{y}\right) = \log_b x - \log_b y$
3. $\log_b(x^r) = r\log_b x$ (where $r$ is any real number)

# Example 6

Use the laws of logarithms to evaluate $\log_2 80 - \log_2 5$.

Solution:

Using Law 2, we have

$$\log_2 80 - \log_2 5 = \log_2 \left(\frac{80}{5}\right)$$

$$= \log_2 16$$

$$= 4$$

because $2^4 = 16$.

# Natural Logarithms

Of all possible bases *b* for logarithms, we will see that the most convenient choice of a base is the number *e*.
The logarithm with base *e* is called the **natural logarithm** and has a special notation:

$$\log_e x = \ln x$$

If we put $b = e$ and replace $\log_e$ with "ln" in (6) and (7), then the defining properties of the natural logarithm function become

**8**

$$\ln x = y \iff e^y = x$$

# Natural Logarithms

9

$$\ln(e^x) = x \qquad x \in \mathbb{R}$$

$$e^{\ln x} = x \qquad x > 0$$

In particular, if we set $x = 1$, we get

$$\ln e = 1$$

# Example 7

Find $x$ if $\ln x = 5$.

Solution 1:

From (8) we see that

$$\ln x = 5 \qquad \text{means} \qquad e^5 = x$$

Therefore $x = e^5$.

(If you have trouble working with the "ln" notation, just replace it by $\log_e$. Then the equation becomes $\log_e x = 5$; so, by the definition of logarithm, $e^5 = x$.)

# Example 7 – *Solution*

Solution 2:

Start with the equation

$$\ln x = 5$$

and apply the exponential function to both sides of the equation:

$$e^{\ln x} = e^5$$

But the second cancellation equation in (9) says that $e^{\ln x} = x$.

Therefore $x = e^5$.

# Natural Logarithms

The following formula shows that logarithms with any base can be expressed in terms of the natural logarithm.

$\boxed{10}$ **Change of Base Formula** For any positive number $b$ $(b \neq 1)$, we have

$$\log_b x = \frac{\ln x}{\ln b}$$

# Example 10

Evaluate $\log_8 5$ correct to six decimal places.

Solution:

Formula 10 gives

$$\log_8 5 = \frac{\ln 5}{\ln 8}$$

$$\approx 0.773976$$

# Inverse Trigonometric Functions

When we try to find the inverse trigonometric functions, we have a slight difficulty: Because the trigonometric functions are not one-to-one, they don’t have inverse functions.

The difficulty is overcome by restricting the domains of these functions so that they become one-to-one.

# Inverse Trigonometric Functions

You can see from Figure 17 that the sine function $y = \sin x$ is not one-to-one (use the Horizontal Line Test).

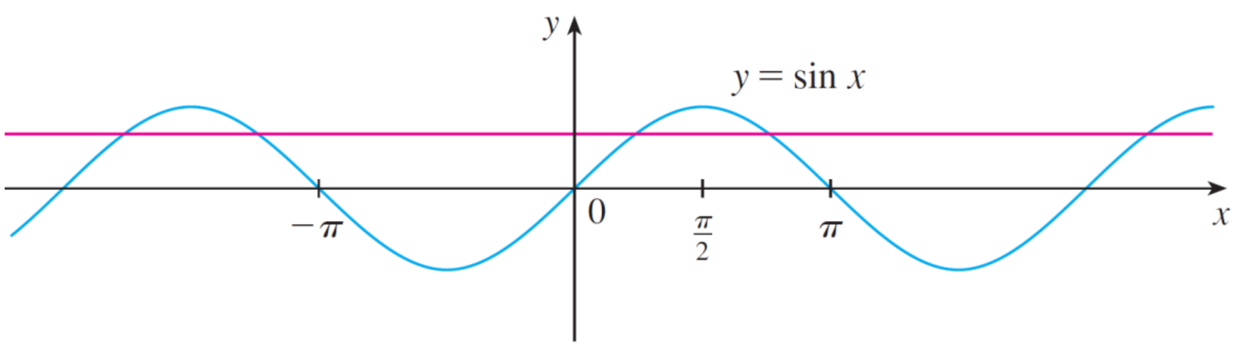
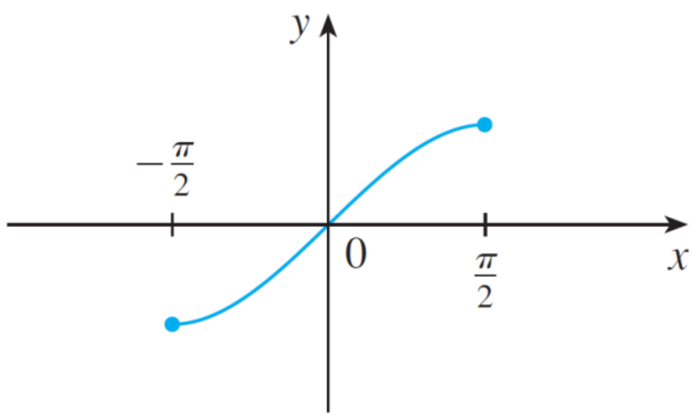



Figure 17

But the function $f(x) = \sin x$, $-\pi/2 \le x \le \pi/2$, *is* one-to-one (see Figure 18).




$y = \sin x, \ -\frac{\pi}{2} \le x \le \frac{\pi}{2}$

Figure 18

# Inverse Trigonometric Functions

The inverse function of this restricted sine function *f* exists and is denoted by $\sin^{-1}$ or arcsin. It is called the **inverse sine function** or the **arcsine function**.

Since the definition of an inverse function says that

$$\Longleftrightarrow$$

$$f^{-1}(x) = y \iff f(y) = x$$

W

$$\sin^{-1}x = y \quad \Longleftrightarrow \quad \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leqslant y \leqslant \frac{\pi}{2}$$

# Inverse Trigonometric Functions

Thus, if $-1 \le x \le 1$, $\sin^{-1}x$ is the number between $-\pi/2$ and $\pi/2$ whose sine is $x$.

# Example 12

Evaluate (a) $\sin^{-1}\left(\frac{1}{2}\right)$ and (b) $\tan(\arcsin \frac{1}{3})$.

Solution:

(a)We have

$$\sin^{-1}\left(\tfrac{1}{2}\right) = \frac{\pi}{6}$$

because $\sin(\pi/6) = \frac{1}{2}$ and $\pi/6$ lies between $-\pi/2$ and $\pi/2$.

# Example 12 – *Solution*

cont'd

(b) Let $\theta = \arcsin \frac{1}{3}$ so $\sin \theta = \frac{1}{3}$

Then we can draw a right triangle with angle $\theta$ as in Figure 19 and deduce from the Pythagorean Theorem that the third side has length $\sqrt{9 - 1} = 2\sqrt{2}$

This enables us to read from the triangle that

$$\tan\left(\arcsin \tfrac{1}{3}\right) = \tan \theta = \frac{1}{2\sqrt{2}}$$

$\sin y = \frac{1}{3}$

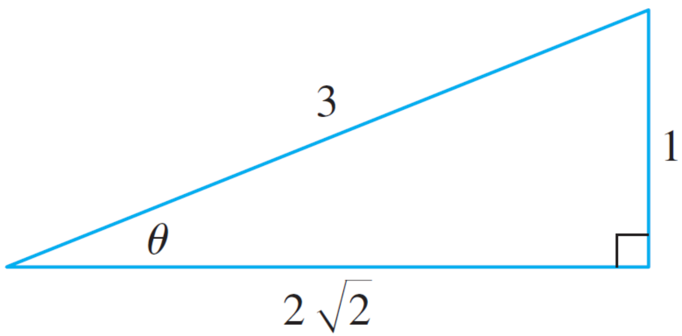



Figure 19

# Inverse Trigonometric Functions

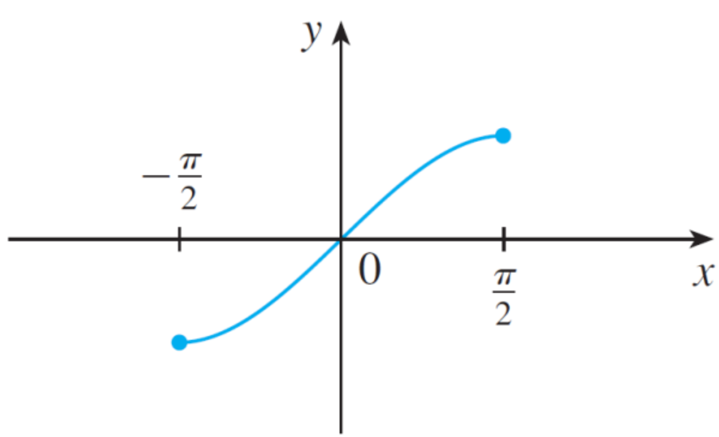
The cancellation equations for inverse functions become, in this case,

$$\sin^{-1}(\sin x) = x \quad \text{for } -\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$$

$$\sin(\sin^{-1} x) = x \quad \text{for } -1 \leqslant x \leqslant 1$$

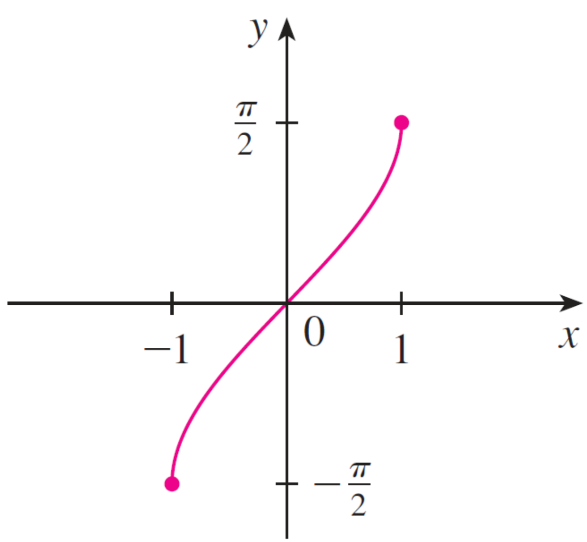
# Inverse Trigonometric Functions

The inverse sine function, $\sin^{-1}$, has domain $[-1, 1]$ and range $[-\pi/2, \pi/2]$, and its graph, shown in Figure 20, is obtained from that of the restricted sine function (Figure 18) by reflection about the line $y = x$.




$y = \sin x, \; -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

**Figure 18**



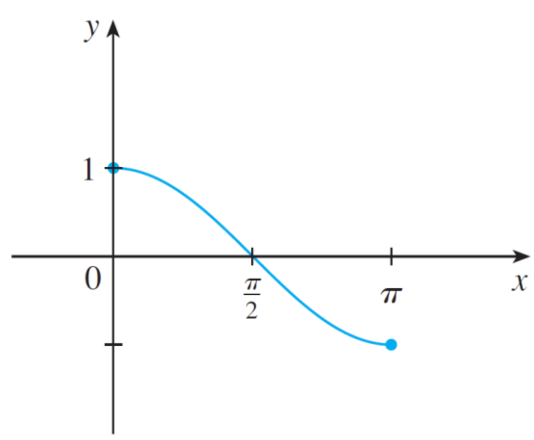

$y = \sin^{-1} x = \arcsin x$

**Figure 20**

# Inverse Trigonometric Functions

The **inverse cosine function** is handled similarly. The restricted cosine function $f(x) = \cos x$, $0 \le x \le \pi$, is one-to-one (see Figure 21) and so it has an inverse function denoted by $\cos^{-1}$ or arccos.

$$\cos^{-1} x = y \iff \cos y = x \quad \text{and} \quad 0 \leqslant y \leqslant \pi$$




$y = \cos x$, $0 \le x \le \pi$

**Figure 21**

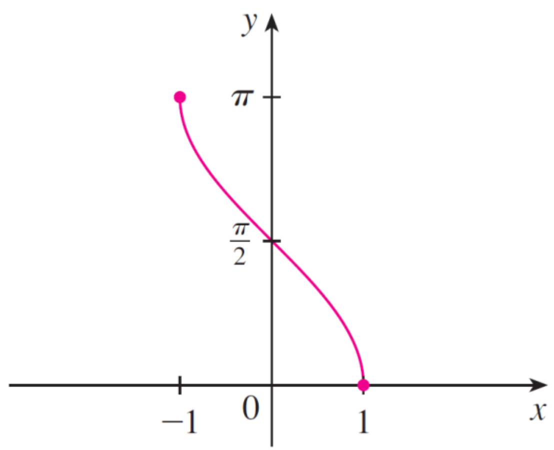
# Inverse Trigonometric Functions

The cancellation equations are

$$\cos^{-1}(\cos x) = x \quad \text{for } 0 \leqslant x \leqslant \pi$$

$$\cos(\cos^{-1}x) = x \quad \text{for } -1 \leqslant x \leqslant 1$$

The inverse cosine function, $\cos^{-1}$, has domain $[-1, 1]$ and range $[0, \pi]$. Its graph is shown in Figure 22.



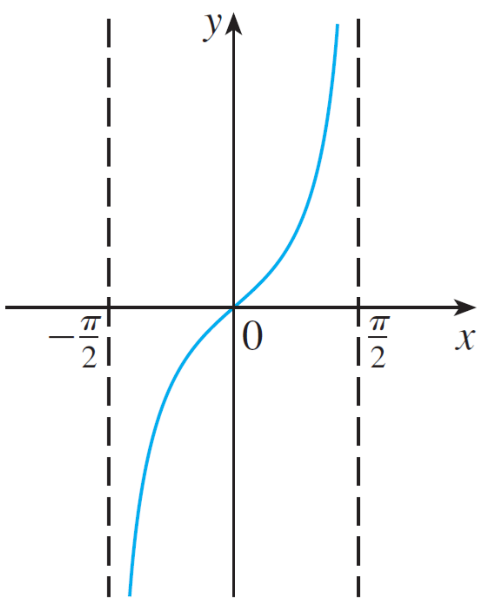

$y = \cos^{-1} x = \arccos x$

Figure 22

# Inverse Trigonometric Functions

The tangent function can be made one-to-one by restricting it to the interval $(-\pi/2, \pi/2)$. Thus the **inverse tangent function** is defined as the inverse of the function $f(x) = \tan x$, $-\pi/2 < x < \pi/2$. (See Figure 23.) It is denoted by $\tan^{-1}$ or arctan.

$$\tan^{-1}x = y \iff \tan y = x \quad \text{and} \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$




$y = \tan x$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$

**Figure 23**

# Inverse Trigonometric Functions

The remaining inverse trigonometric functions are summarized here.

**11**

$$y = \csc^{-1}x \ (|x| \geqslant 1) \iff \csc y = x \quad \text{and} \quad y \in (0, \pi/2] \cup (\pi, 3\pi/2]$$

$$y = \sec^{-1}x \ (|x| \geqslant 1) \iff \sec y = x \quad \text{and} \quad y \in [0, \pi/2) \cup [\pi, 3\pi/2)$$

$$y = \cot^{-1}x \ (x \in \mathbb{R}) \iff \cot y = x \quad \text{and} \quad y \in (0, \pi)$$